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(54) **PLASTIC NAIL EXTENSION AND USE IN A NAIL APPLIQUE KIT**

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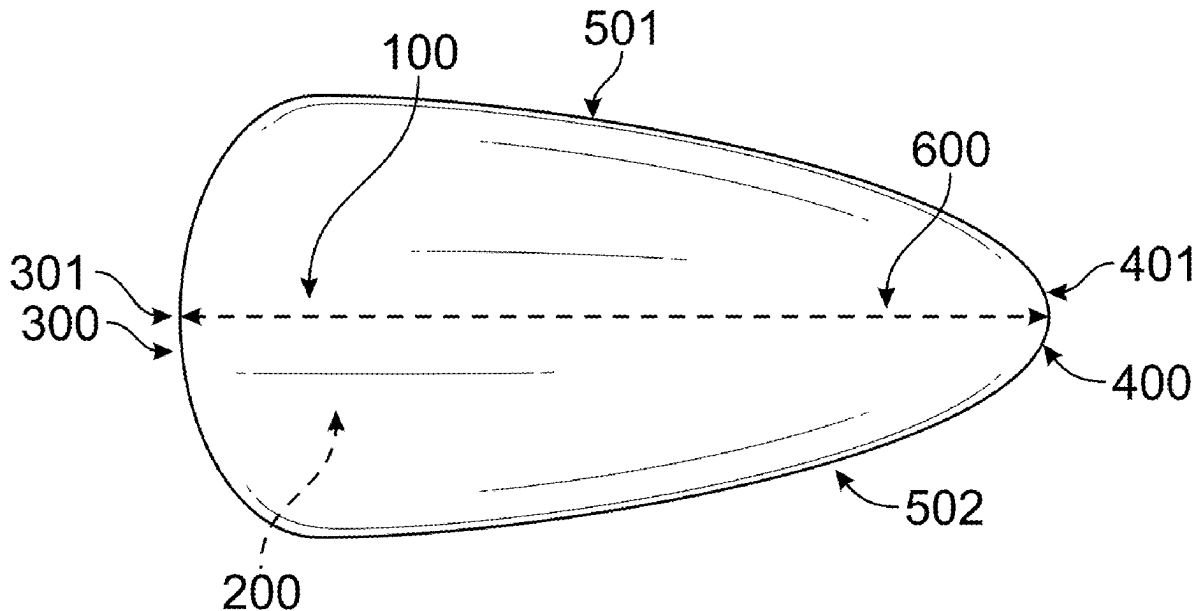
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Related U.S. Application Data

(60) Provisional application No. 63/562,766, filed on Mar. 8, 2024.

(57) **ABSTRACT**

The present invention is directed to a plastic extension for a nail covering and a nail applique kit including at least one plastic extension, an adhesive including a UV gel a cyanoacrylate, a free radical curable monomer composition or a PSA and an optional processing aid.



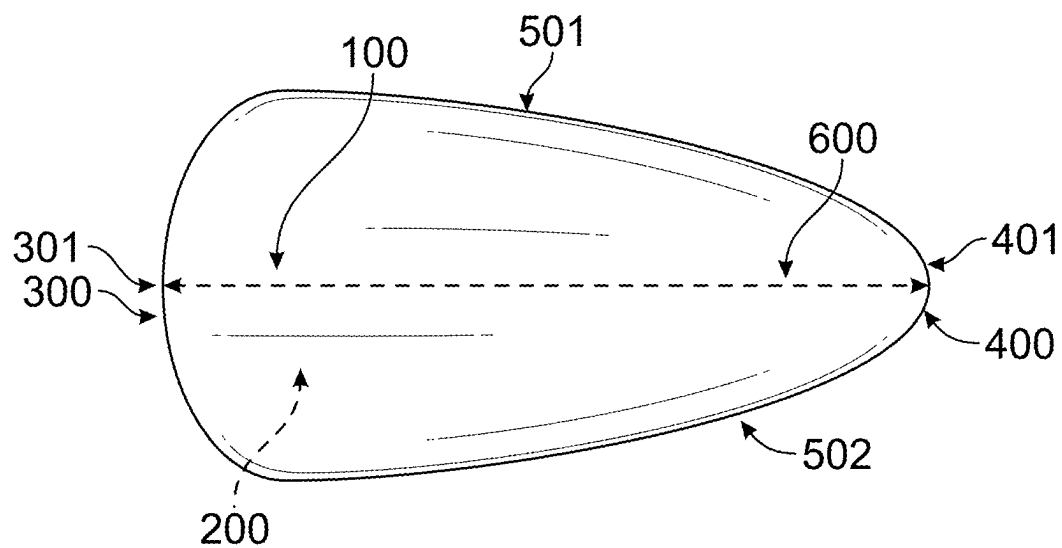


FIG. 1

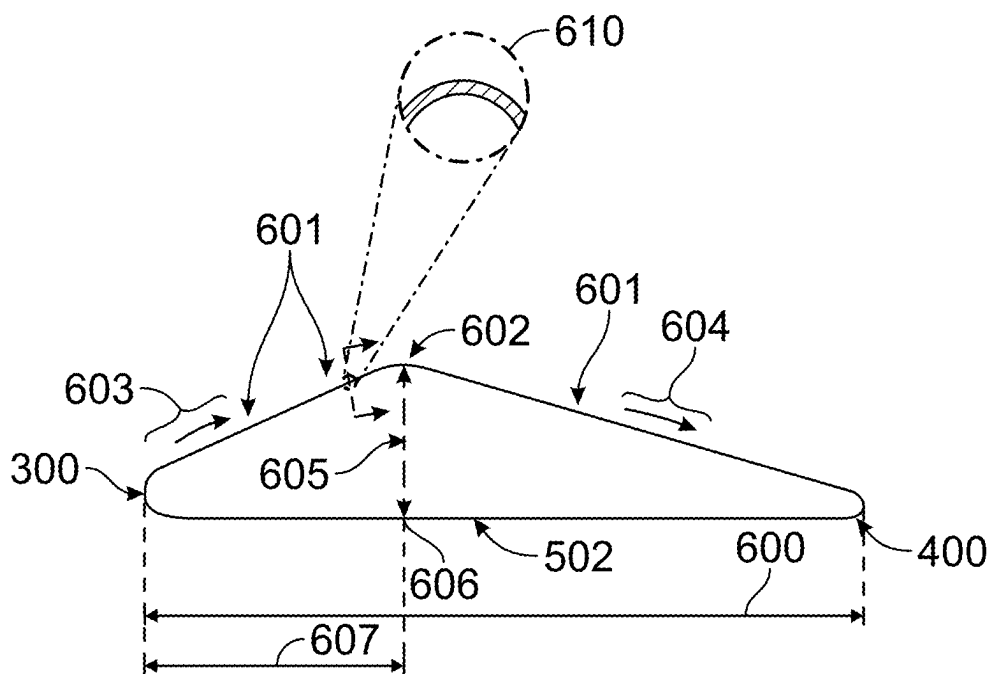


FIG. 2

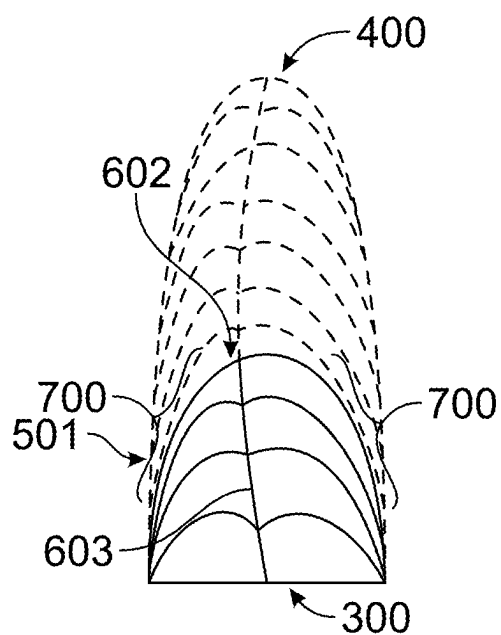


FIG. 3A

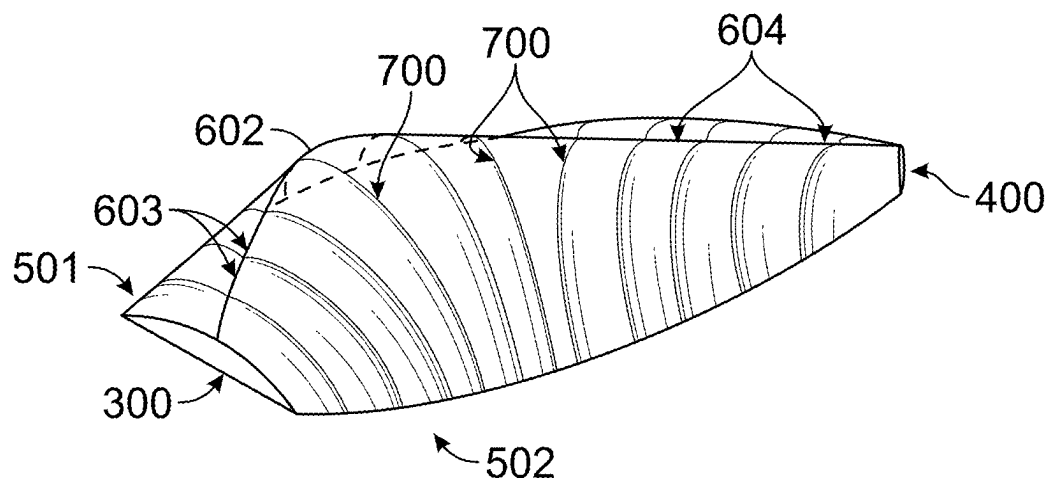


FIG. 3B

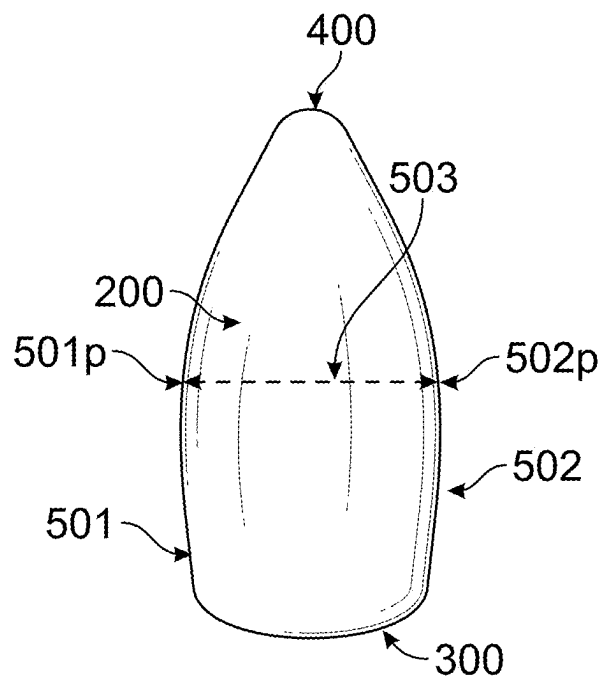


FIG. 4

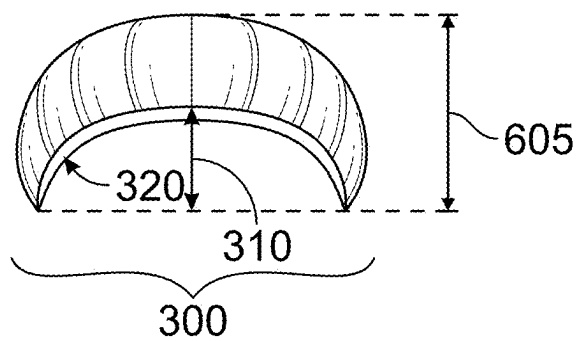


FIG. 5A

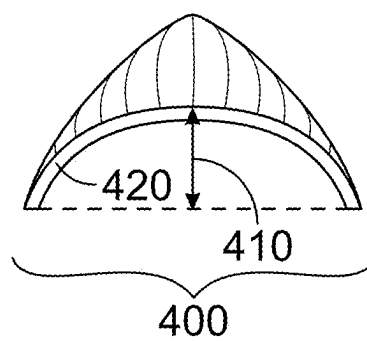


FIG. 5B

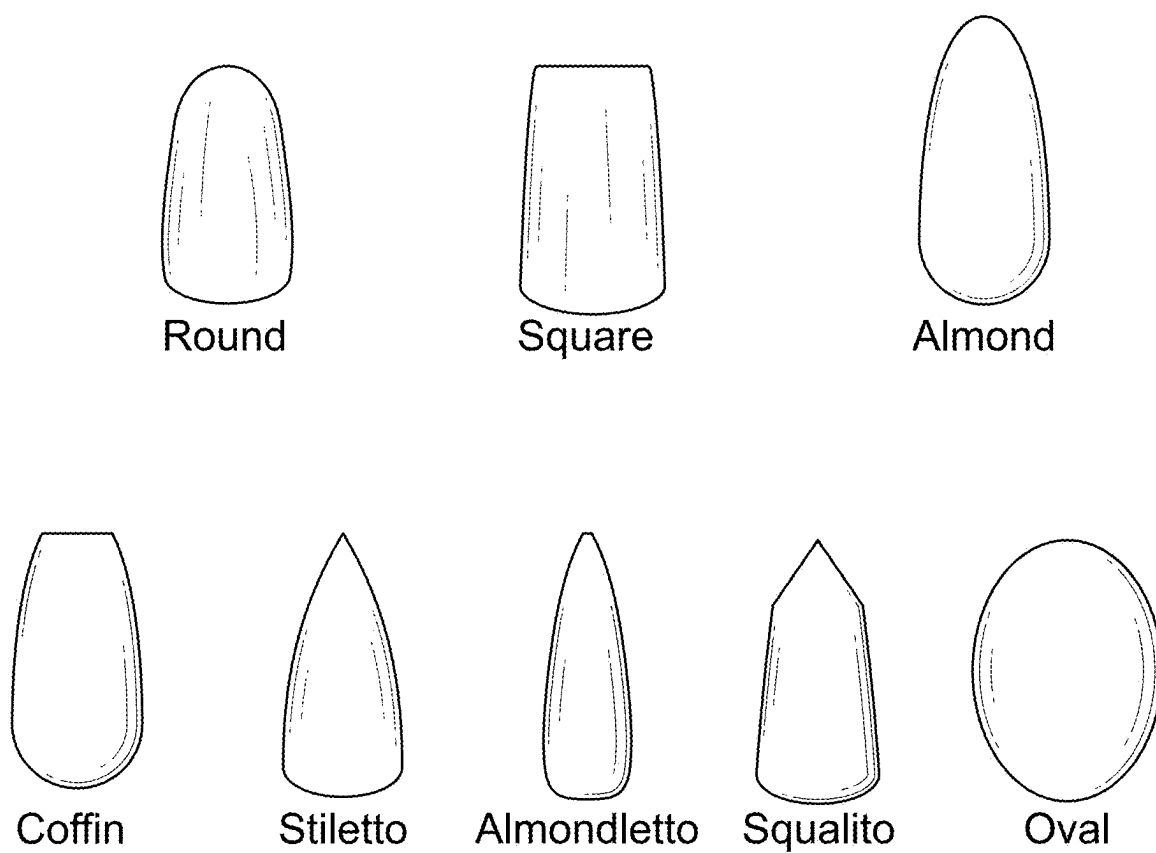


FIG. 6

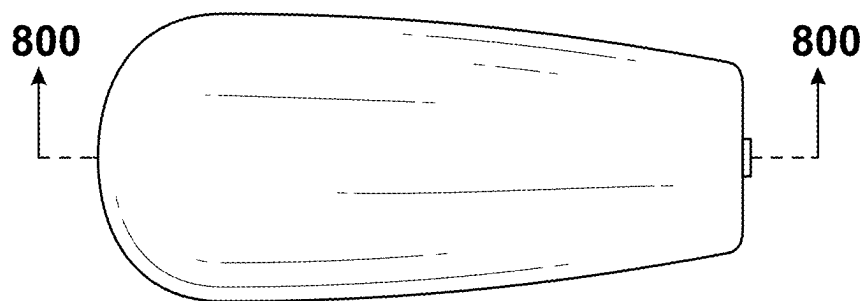


FIG. 7A

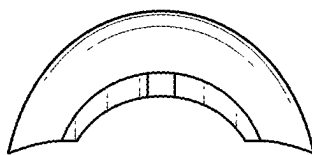


FIG. 7B



FIG. 7C

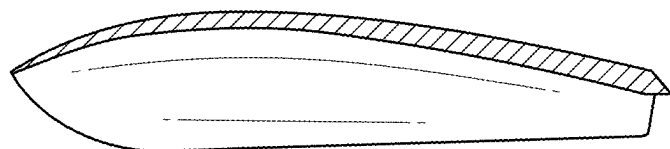


FIG. 7D

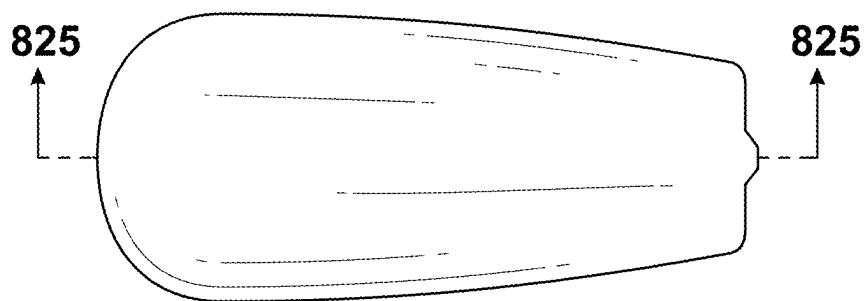


FIG. 8A



FIG. 8B

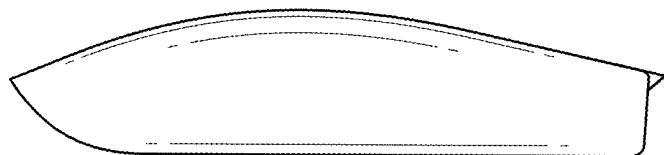


FIG. 8C

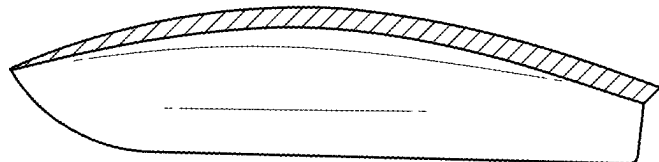


FIG. 8D

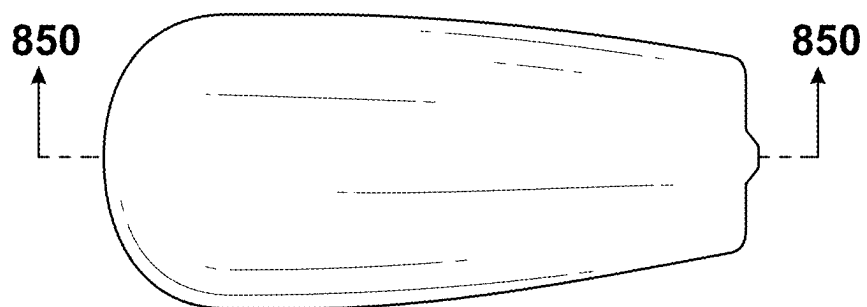


FIG. 9A



FIG. 9B

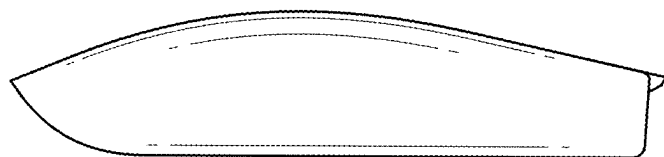


FIG. 9C

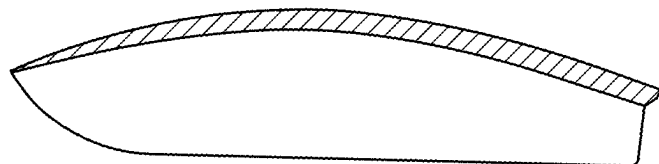


FIG. 9D

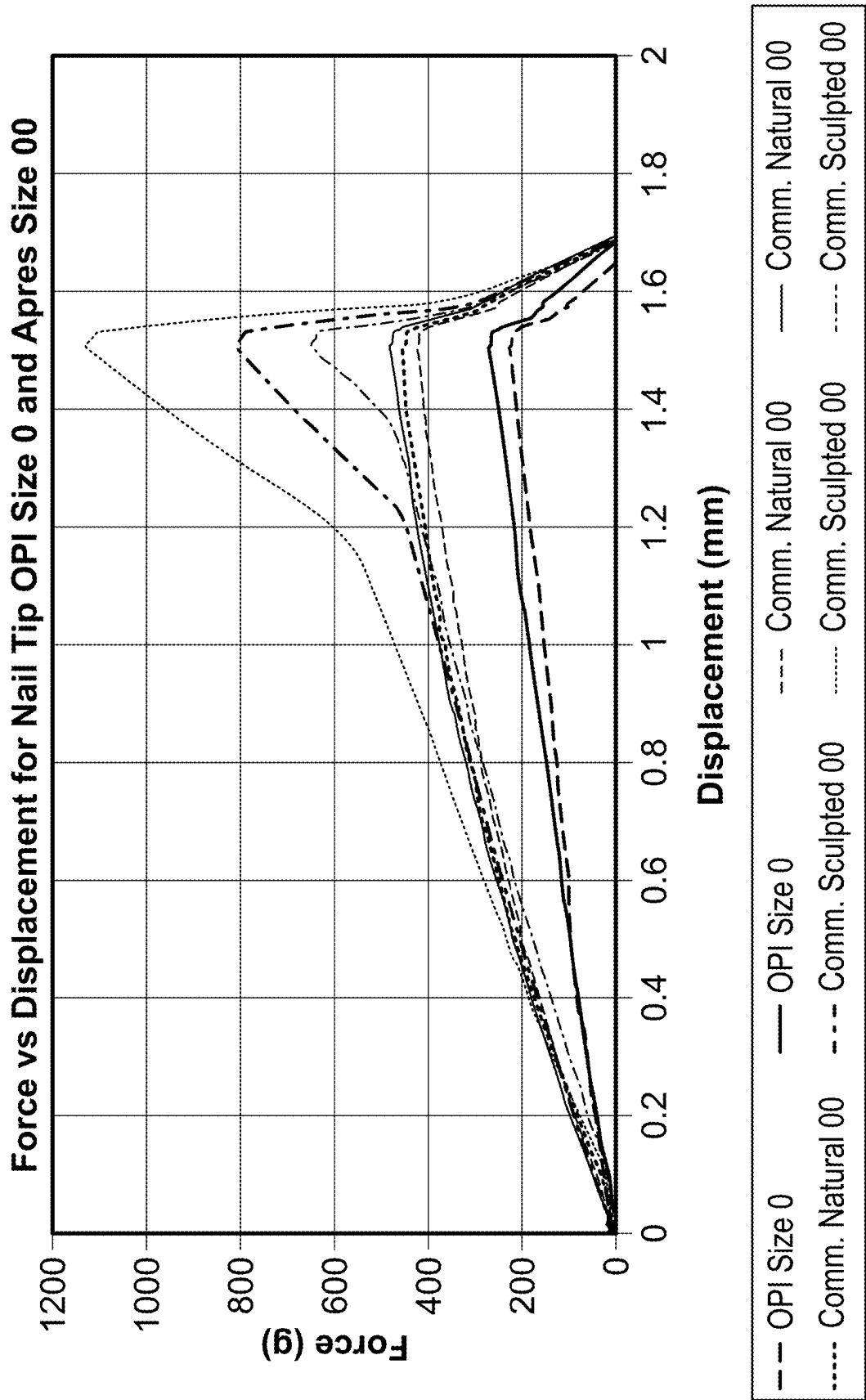


FIG. 10A

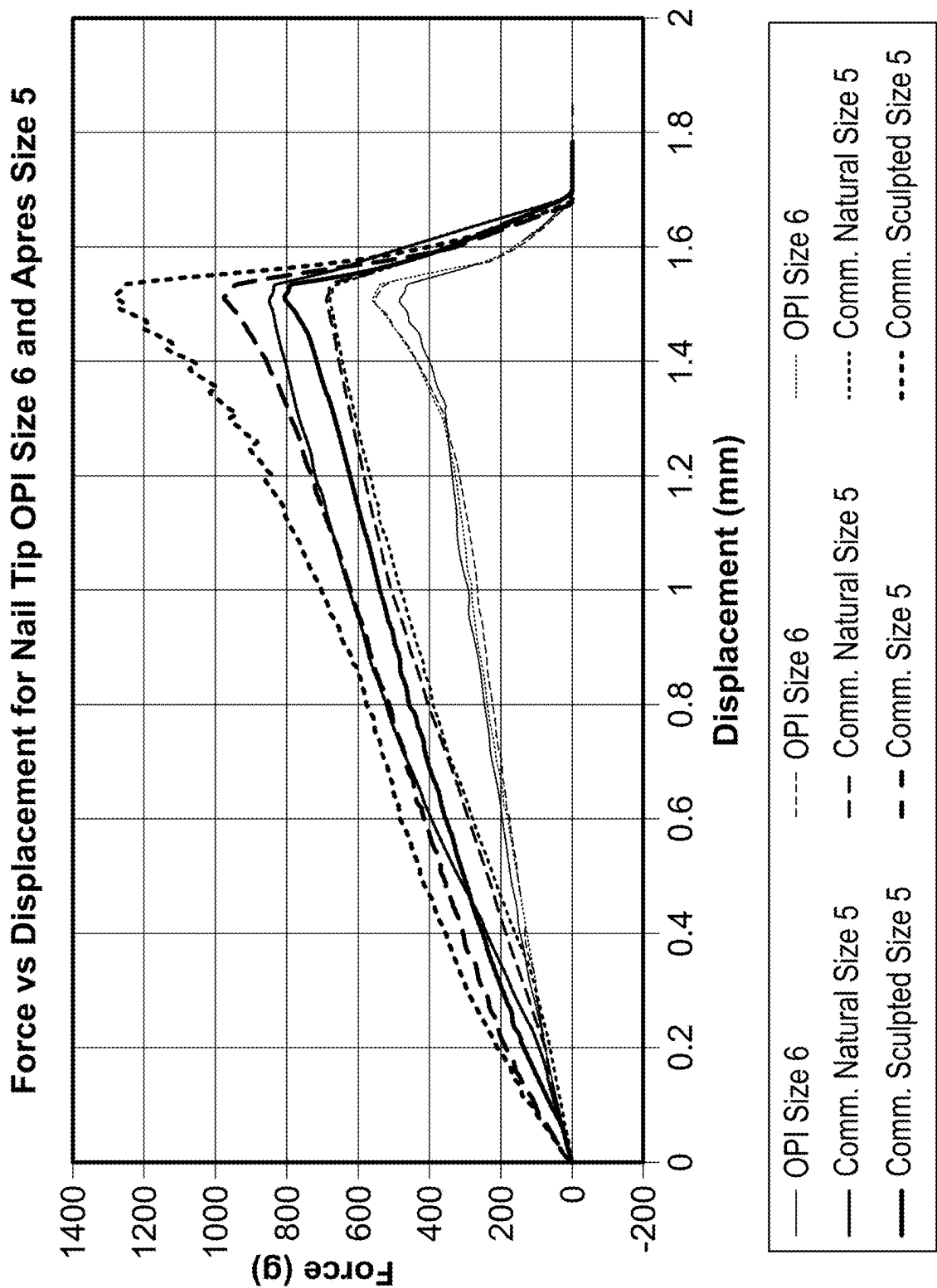


FIG. 10B

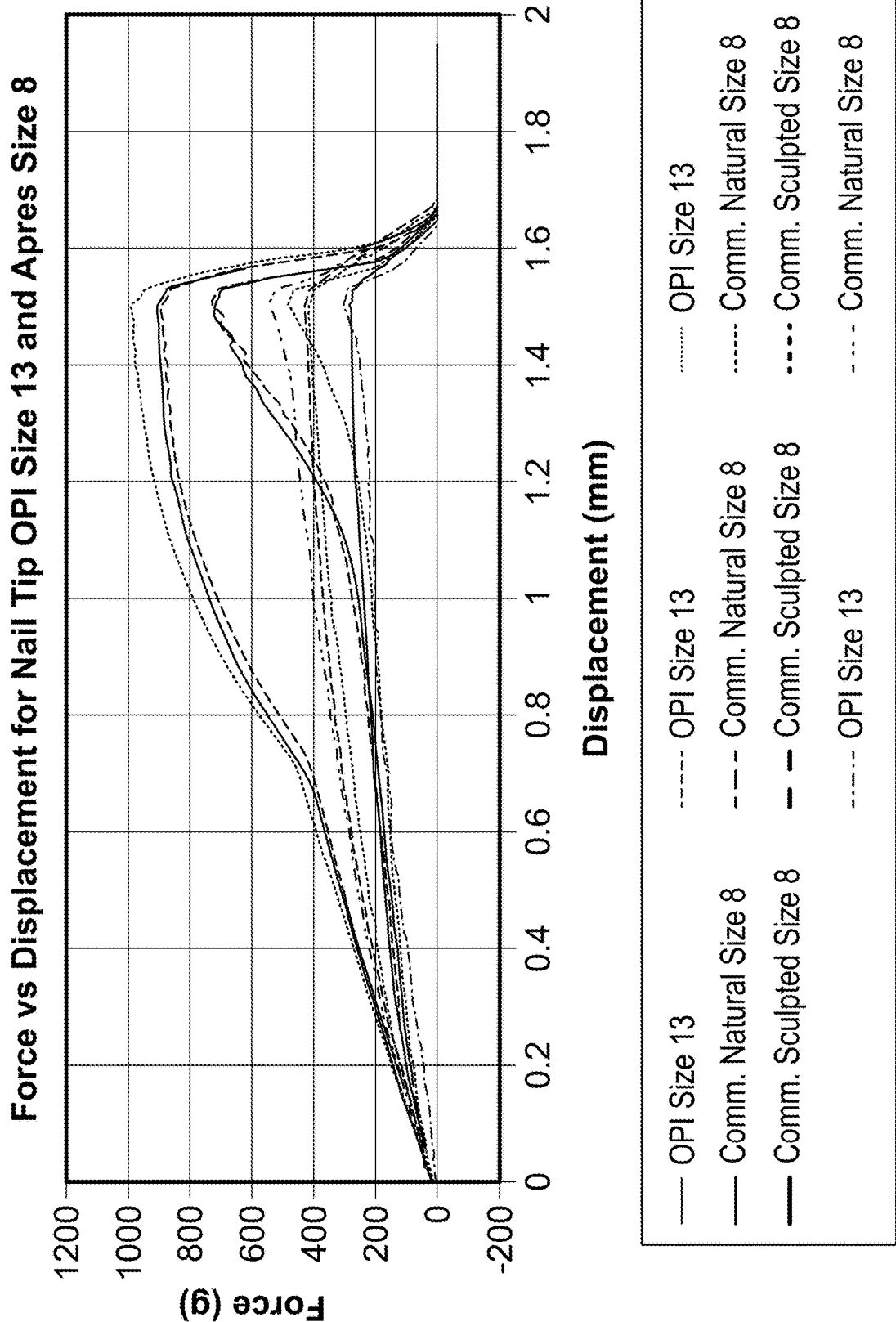
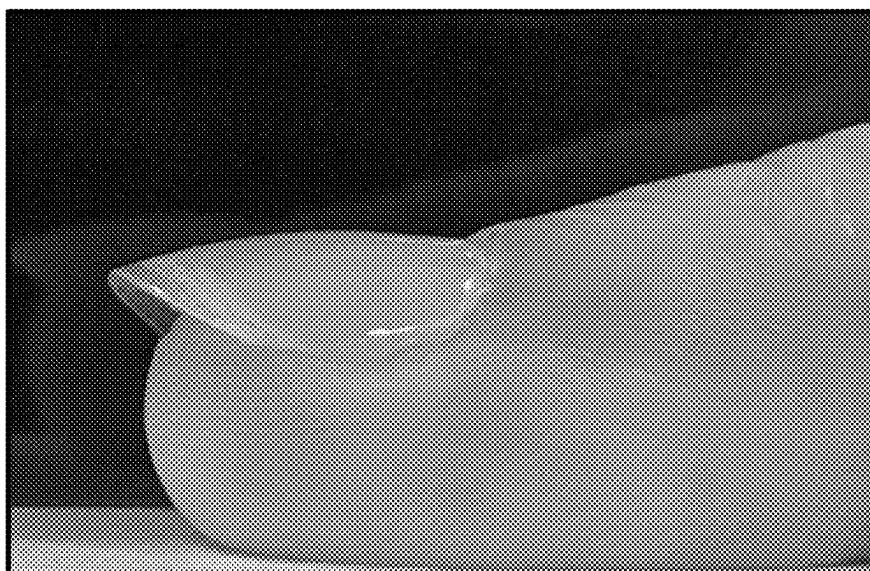


FIG. 10C



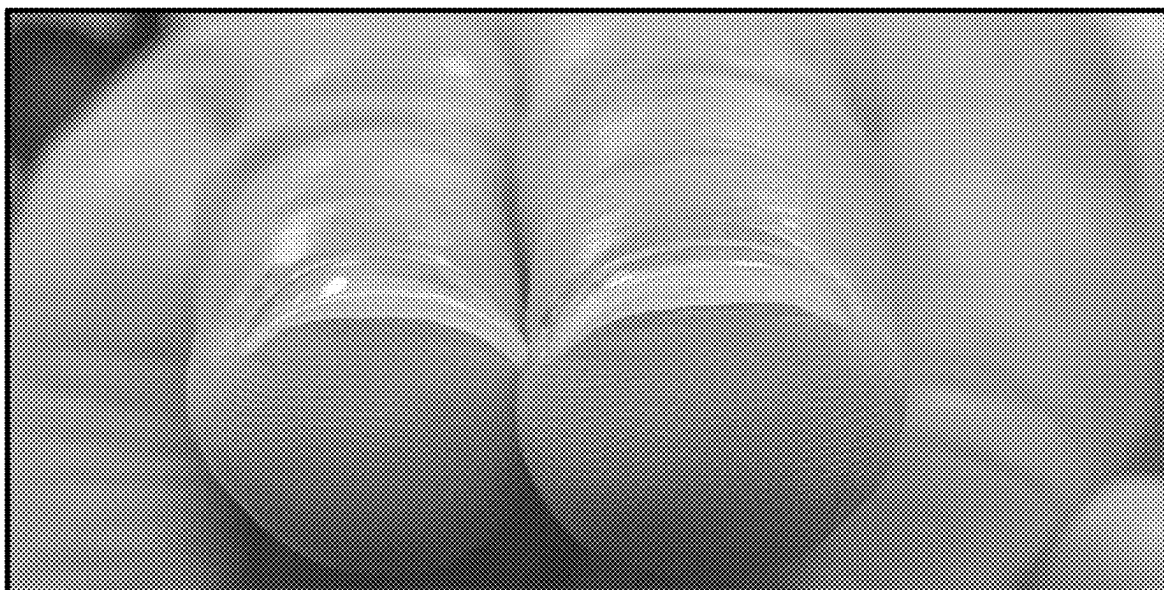
Middle Finger

FIG. 11A



Ring Finger

FIG. 11B



Ring Finger

Middle Finger

FIG. 12

PLASTIC NAIL EXTENSION AND USE IN A NAIL APPLIQUE KIT

CLAIM OF PRIORITY

[0001] This patent application claims the benefit of priority to U.S. Provisional Application Ser. No. 63/562,766, filed Mar. 8, 2024, which is incorporated by reference herein in its entirety.

FIELD OF INVENTION

[0002] The present invention is directed to embodiments of a plastic extension for application to a fingernail and embodiments of an applique kit with one or more plastic extension embodiments, an adhesive and an optional processing aid. The present invention is directed as well to embodiments of a method using the kit for fitting the plastic extension embodiments onto nail plates and the constructs of the resulting plastic extensions bonded to fingernails.

BACKGROUND

[0003] The advent of nail stickers and appliques began with the 600 BC appearance of “Blinged-Out” gold and jewel coverings of the lengthy nails of aristocratic Chinese. The global awareness of the beauty surrounding these kinds of nail paintings and coverings slowly expanded through the trade routes to become a Victorian practice during the 1800’s through use of stain treatment of nails with lightly colored oils. Recognition of such treatments was memorialized in such writings as Flaubert’s “Madame Bovary” novel. In the early 1900’s, societal views of nail coverings and paintings grew with societal change so that fancy coloring of nails became in vogue. Coverings (e.g., stickers) for treatment of broken or damaged nails arose at the same time and used paper residues from coffee filters and tea bags. The protein and carbohydrate substances embedded in the paper residues served as a glue to hold the papers in place on the nails (and perhaps help treatment).

[0004] Nail Wrap kits appeared on the market in the ‘60’s with the Juliette Manicure wrap. This wrap incorporated a small, shaped sheet similar to parchment paper. At about the same time, cigarette papers reinforced with airplane glue began their appearance as nail coverings. These coverings were cut to shape after application to nails. They were problematic due to the imprecise fitting to the nails and the tendency of the edges to peel. Silk and linen replaced the cigarette papers shortly thereafter and provided strength when undercoated with natural adhesive such as casein glue, hide glue or albumen glue.

[0005] The difficulties associated with forerunner single layer nail wraps and appliques on the market since the 1920’s led to the development of the more complex appliques. These appliques have been constructed of single layer materials with adhesive backings. This single layer construction has been based upon two very different kinds of appliques:

- [0006] a) flexible appliques of paper or preformed elastic polymer and,
- [0007] b) inflexible appliques of shaped solid inelastic plastic.

[0008] While convenient to use and easily fitted to nail contours, the flexible paper and preformed elastic polymer appliques have performance difficulties including the need to trim after fitting and the tendency to wrinkle and fray.

These kinds of appliques can be easily displaced from nails because of tearing, edge fragmenting and similar disadvantages owing to their thin fragile construction.

[0009] The inflexible solid plastic appliques on the other hand are multiply pre-sized to fit different sized nails so that trimming is usually not needed. They are constructed of solid, shaped plastic pieces of ABS, Nylon, PMMA, PVC, PST or PC polymer. They are press/heat formed to have concave/convex shapes matching the contours and outlines of fingernails. The plastic pieces have on their undersides PSA adhesive coatings. However, even though they are sold in a range of sizes, these inflexible appliques do not fit precisely onto individual nail contours because individual nail contours do not follow a uniform contour. The thicker sizes of the plastic appliques relative to the very thin flexible paper appliques lead to a perception of discomfort in use. The polymeric material of the plastic appliques either is so hard that it is difficult to remove or is so soft so that it tends to fray and crack. Moreover, the PSA adhesive coupled with the thick plastic of these plastic appliques frequently causes “pop-off” of the applique especially when an edge of the applique is caught by a table, a chair or clothing.

[0010] For these reasons, developments of plastic appliques and associated application methods that avoid and/or minimize these problems are sought. Among the objects for these developments are appliques that do not have “pop-off” problems and appliques that provide a comfortable feeling when attached to fingernails.

SUMMARY

[0011] These and other objects are achieved by aspects of the present invention which are directed to embodiments of a plastic extension for fingernails that prevent and/or minimize “pop-offs.” The embodiments of the plastic extension provide construction and parameters that deliver significant comfort and avoid an “artificial appearance.” Embodiments of the plastic extension are also components of a corresponding applique kit.

[0012] Embodiments of the plastic extension also constitute parts for embodiments of the method of the invention that enable development of a firmly held, comfortable, attractive plastic extension on a nail plate.

[0013] Embodiments of the invention incorporating these features include the applique kit comprising at least a two-part combination of the plastic extension embodiment and an adhesive and preferably a three part combination which adds to the two part combination a processing aid for managing the adhesive. The adhesive may be a UV curable gel, a cyanoacrylate adhesive, a free radical curable composition, or a pressure sensitive adhesive. Embodiments of a preferred applique kit are the three part applique kit including embodiments of the plastic extension, a UV curable gel and a UV LED lamp.

[0014] Embodiments of the plastic extension according to the invention incorporate variations of shape and construction compared with commercial plastic appliques so that the embodiments achieve improvements in fit, comfort, wearability, longevity, appearance and shape on the nails. Embodiments of the method of attachment of the plastic extension to a nail plate achieve protection of the nail plate, elimination and/or minimization of pop-off, avoidance of unacceptable and/or potentially harmful chemicals, ease of

manipulation of attachment, avoidance of unacceptable sensory perceptions and development of pleasing sensory perceptions.

[0015] Embodiments of the plastic extension include features for the cuticle and distal ends, left and right sides, top and bottom surfaces and widths and lengths. Embodiments of the plastic extension are characterized by their length categories and their size categories. The length categories include long, medium and short. The size categories are the widths of the plastic extension for fitting the width of natural nails.

[0016] The embodiments of the plastic extension according to the invention are curved in three dimensions provided by intersecting longitudinal and latitudinal curves which run the lengths and widths of the plastic extension embodiments respectively. The longitudinal curves for the long, medium and short length categories have apexes located at certain ratios of the plastic extension lengths as specified in the Detailed Description. The distance of the apex position relative to the cuticle end is the apex length. Partitioned according to length categories, long medium and short, is a ratio of the apex length to the length of the plastic extension. Included also is a ratio of the cuticle height to apex height.

[0017] Embodiments of the plastic extension do not rock or move when placed on a flat surface and a downward force is applied to either the cuticle end or the distal end. The combination of parameters for the cuticle thickness, the apex thickness, the ratio of apex length to total length, the apex position, the ratio of cuticle height to apex height, and low cuticle end flex resistance enable facile attachment of the cuticle end, apex and distal end of embodiments of the plastic extension to natural nails and provide a glove fit of the embodiments onto the fingernail plates. These features deliver comfortable, fitted plastic extension embodiments onto fingernails and provide attractive natural shapes. These features enhance fit, comfort, and wearability of the plastic extension embodiments. These features also enhance the attractiveness of the plastic extension embodiments attached to nail plates.

BRIEF DESCRIPTIONS OF THE DRAWINGS

[0018] FIG. 1 depicts a top plane view of the plastic extension.

[0019] FIG. 2 depicts a side plane view of the plastic extension.

[0020] FIG. 3A depicts an elevated cuticle end view of the plastic extension.

[0021] FIG. 3B depicts a top perspective view of the plastic extension.

[0022] FIG. 4 depicts a bottom view of the plastic extension.

[0023] FIG. 5A depicts a cuticle end plane view of the plastic extension.

[0024] FIG. 5B depicts a distal (tip) end plane view of the plastic extension.

[0025] FIG. 6 depicts the eight typical overall shapes of the plastic extension.

[0026] FIG. 7A depicts a digital scan photograph of the top view.

[0027] FIG. 7A depicts a digital scan photograph of the top view of an embodiment of the plastic extension.

[0028] FIG. 7B depicts a digital scan photograph of the cuticle end view of the embodiment of the plastic extension.

[0029] FIG. 7C depicts a digital scan photograph of the side view of the embodiment of the plastic extension.

[0030] FIG. 7D depicts a digital scan photograph of a cross-sectional thickness of an embodiment of the plastic extension along a line 800 in FIG. 7A.

[0031] FIG. 8A depicts a digital scan photograph of the top view of a first commercial plastic nail tip.

[0032] FIG. 8B depicts a digital scan photograph of the cuticle end view of the first commercial plastic nail tip.

[0033] FIG. 8C depicts a digital scan photograph of the side view of the first commercial plastic nail tip.

[0034] FIG. 8D depicts a digital scan photograph of a cross-sectional thickness of a first commercial plastic nail tip along a line 825 in FIG. 8A.

[0035] FIG. 9A depicts a digital scan photograph of the top view of a second commercial plastic nail tip.

[0036] FIG. 9B depicts a digital scan photograph of the cuticle end view of the second commercial plastic nail tip.

[0037] FIG. 9C depicts a digital scan photograph of the side view of the second commercial plastic nail tip.

[0038] FIG. 9D depicts a digital scan photograph of a cross-sectional thickness of the second commercial plastic nail tip along a line 850 in FIG. 9A.

[0039] FIGS. 10A, 10B and 10C are graphs of the flex resistance over time for embodiments of the plastic extensions of the invention and commercial nail tips.

[0040] FIGS. 11A and 11B present photographs of a commercial tip attached to the middle finger of the test subject and an individual plastic extension attached to the ring finger of the test subject.

[0041] FIG. 12 presents a photograph of the ends of the middle finger with commercial tip attached and ring finger with plastic extension attached as shown in FIGS. 11A and 11B.

DEFINITIONS OF AND FOR THE INVENTION

[0042] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by a person of ordinary skill in the art.

[0043] As used in the specification and the appended statements and claims, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Also, use of a plural term describing a thing or element includes the singular unless the context clearly dictates otherwise. For example, the term “embodiments of a plastic extension” includes a single embodiment of a plastic extension as well as multiple embodiments of a plastic extension.

[0044] The term “may” in the context of this application means “is permitted to” or “is able to” and is a synonym for the terms “can” and “is.” The term “may” as used herein does not mean possibility or chance.

[0045] The term and/or in the context of this application means one or the other or both. For example, an aqueous solution of A and/or B means an aqueous solution of A alone, an aqueous solution of B alone and an aqueous solution of a combination of A and B. When more than two items are referred, the term and/or also means any combination of these multiple items as well as all and each.

[0046] The terms “about” and “approximately” as used herein, when referring to a numerical value or range, allow for a degree of variability in the value or range, and are understood to mean

[0047] +10%, of a stated value or of a stated limit of a range.

[0048] If a value of a variable that is necessarily an integer, e.g., the number of carbon atoms in an alkyl group or the number of substituents on a ring, is described as a range, e.g., 0-4, what is meant is that the value can be any integer between 0 and 4 inclusive, i.e., 0, 1, 2, 3, or 4. Similarly, values expressed in a range format should be interpreted in a flexible manner to include not only the numerical values explicitly recited as the limits of the range, but also to include all the individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range were explicitly recited. For example, a range of "about 0.1% to about 5%" or "about 0.1% to 5%" should be interpreted to include not just about 0.1% to about 5%, but also the individual values (e.g., 1%, 2%, 3%, and 4%) and the sub-ranges (e.g., 0.1% to 0.5%, 1.1% to 2.2%, 3.3% to 4.4%) within the indicated range.

[0049] The terms "comprising", "comprises" and "comprised of" as used herein are synonymous with "including", "includes" or "containing", "contains", and are inclusive or open-ended and do not exclude additional, non-recited members, elements or method steps.

[0050] For the purposes of the presently claimed invention, "% by weight" or "wt %" as used in the presently claimed invention is with respect to the total weight of the recited composition. Further, the sum of wt % of all the compounds (components), as described herein, in the respective compositions may add up to 100 wt % when optional non-delineated components such as solvent, plasticizer, film formers, ancillary additives and the like are included. Accordingly, all percents of ingredients or components are given as weight-percentages relative to the total weight of the composition in which the ingredients and/or components are present, unless otherwise stated.

[0051] All average molecular weights of polymers are weight-average molecular weights, unless otherwise specified.

[0052] The term "substantially free" as the term is used herein means completely or almost completely; for example, a composition that is "substantially free" of a component either has none of the component or contains such a small amount that any relevant functional property of the composition is unaffected by the presence of the small amount of the component in question. A compound that is "substantially pure" has only negligible traces of impurities present.

[0053] The term "substantial" means a significant amount such as more than a majority amount. For example, a mixture of compounds A and B in which A is present in a substantial amount means that A is present at a weight percent or number of moles that is greater than the weight percent or number of moles of B. This term also means more than a minimal characteristic, examples of which include substantial flow or substantial transformation.

[0054] The terms "essential" and "essentially" are encompassed by the terms substantial and substantially so as to cover a smaller range than the terms substantial and substantially. The terms essential and essentially relating to purity or result equal to or more than 98% of the theoretic 100% purity or result.

[0055] The following groups of terms are used throughout this application: 1) preferred, preferably and preferable; 2) more preferred, more preferably and more preferable, 3) especially more preferred, especially more preferably and

especially more preferable; 4) most preferred, most preferably and most preferably; 5) especially most preferred, especially most preferably and especially most preferable; and 6) very especially most preferred, very especially most preferably and very especially most preferable. These groups convey a meaning of preference for a group of substituents, structures, moieties, components and compounds. The degree of preference is self-explanatory by the terms themselves. Within each group, the meanings of the synonyms, preferred, preferably and preferable are the same. There is no difference in meaning in the context of this application when a group is described in a particular sentence as preferred and then in another sentence this same group is described as preferably. Not all six categories of preference are used in this application to describe each and every substituent, formula, subgenus integer symbol and atom designator. In some instances, two or three categories are used while in other categories five or six categories are used. The degree of preference as expressed by these terms for members of series which progress from many to a few individually named components is self-explanatory and internally consistent for the particular series being described.

[0056] Reference throughout this specification to "embodiment", "one embodiment" or "preferred embodiment" or "more preferred embodiment" or "most preferred embodiment" means a feature of any one or more elements and can be used in connect with different elements of the invention.

[0057] The term film former means a fully formed polymer such as a poly(meth)acrylate, polyester, polyamide, polyurethane, cellulosic ether or ester or shellac that will form a film or layer when the film former in an organic or aqueous medium is coated on a substrate and dried to remove the medium. The film former typically does not undergo substantial cross-linking or other chemical reaction after its deposition as a film.

[0058] The term (meth)acrylate means either one alone and/or both of an acrylate ester and a methacrylate ester. When an alkyl methacrylate ester alone is to be described, the parenthesis around "meth" is excluded and the resulting term means the 1-methyl-1-carbanoyloxyalkyl ethene or alkyl methacrylate. In a similar fashion, the term alkyl acrylate means 1-carbanoyloxyalkyl ethene or alkyl acrylate.

[0059] The term monomer means an olefinic unsaturated organic compound, a.k.a. olefinic monomer, that is capable of undergoing polymerization to form a polymer having a carbon backbone. As applied to a (meth)acrylate monomer, this term means a (meth)acrylate monomer that is capable of polymerizing to form a poly (meth)acrylate.

[0060] The term dimer means an organic compound having at least two olefinic bonds, a.k.a. olefinic dimer, that is capable of chain extension and chain crosslinking when combined with an olefinic monomer.

[0061] The term surfactant means a zwitterionic, nonionic, anionic or cationic compound having lipophilic and hydrophilic qualities so that it can function to solubilize lipophilic substances in hydrophilic and/or aqueous media. A surfactant may also perform as a plasticizer in polymer compositions to provide flexibility to the polymer composition.

[0062] The term plasticizer means a compound that may be soluble to dispersible in a solid material such as a polymer and enables the macromolecular configuration of

the mixture to exhibit flexibility not present in the polymer alone often by lowering the glass transition temperature curve of the polymer. Plasticizers for polyolefins and functionalized polyolefins are well-known. Examples include alkyl phthalate esters, alkyl adipate esters, alkyl sebacate esters, glycerol triacetate (triacetin) and acetyl tributyl citrate.

[0063] The term rheologic control agent means a rheology modifier that will thicken an otherwise free-flowing liquid composition. The modification renders the composition flowable when applied by spray, brush or other coating method but the composition will remain in a static position in a quiescent state. These agents are typically classed as thixotropic agents. Examples include polyvinyl alcohol, a polyethylene glycol, vegetable gum, laponites, smectite or montmorillonite clays optionally semisynthetically modified and hydrocarbon wax.

[0064] The term alcohol means a mono, di, tri or polyol that can solubilize other components of the composition. Alcohols include methanol, ethanol, propanol, butanol, ethylene glycol (ethylene diol), glycerin (glycerol), propylene diol, or di-ethylene glycol, di-propylene glycol as well as glyme (methoxylated ethylene glycol). Preferred alcohols include but are not limited to methanol, ethanol, propanol, ethylene glycol, propylene glycol and glycerin.

[0065] The term gel means a liquid within a three-dimensional network that forms at least a semi-solid-like consistency in the static, quiescent state. The gel has sufficient rheologic control and/or density and/or viscosity at rest (static state) to maintain continuous integrity of the liquid gel as a coating or layer on a flat or curved surface. The gel character of the liquid means that the liquid will not spontaneously flow off a surface on which it has been coated but can be readily removed by mechanical force such as by wiping with a cloth or tissue.

[0066] The term flowable means a liquid that will flow like water or an aqueous latex paint when contacted by mechanical means such as by a brush, sponge or other applicator.

[0067] Together, the terms gel and flowable mean that the liquid having these characteristics is thixotropic.

[0068] The terms nail or fingernail or natural nail mean a human fingernail and/or a human toenail and/or a human thumbnail having a cuticle end located where the nail emerges from under the skin (proximal nail fold) of the finger, thumb or toe, a distal end located at the end of the nail opposite the cuticle end, and a smile line which is the curved line of the nail where the nail ends its connection with the nail bed and the nail free edge starts. The term "nail plate" is synonymous with nail or fingernail. All of these terms refer to the keratinaceous horny plate (nail) on top of the flesh forming the nail bed.

[0069] The term "size" in relation to plastic fingertips and appliques including but not limited to plastic extension embodiments according to the invention means width and does not include length. According to convention, there are sixteen size categories numbered and half numbered from Double-zero (DO) to zero (0) to 9.

[0070] The term "user" means the person preparing and applying an artificial nail extension of the present invention. The user may be, for example, a professional manicurist who is different from the person on whose nails the artificial nail extension is to be applied. The user, for example, may also be the same as the person receiving the artificial nail extensions, i.e., the user may be the consumer.

[0071] In the following passages, different aspects of the subject matter are defined in more detail. Each aspect so defined may be combined with any other aspect or aspects unless clearly indicated to the contrary. In particular, any feature indicated as being preferred or advantageous may be combined with any other feature or features indicated as being preferred or advantageous. A particular feature, structure, or characteristic described in connection with an embodiment may not be the same as another feature, structure or characteristic of another embodiment of the presently claimed invention. Thus, appearances of the phrases "in one embodiment" or "an embodiment" or "in a preferred embodiment" in various places throughout this specification are not necessarily all referring to the same embodiment but may refer to different embodiments of the presently claimed invention. Furthermore, the features, structures, or characteristics in one or more embodiments may be combined in any suitable manner, as would be apparent to a person skilled in the art from this disclosure. Furthermore, while some embodiments described herein include some, but not other features included in other embodiments, combinations of features of different embodiments are meant to be within the scope of the subject matter, and form different embodiments, as would be understood by those skilled in the art. For example, in the appended claims, any of the claimed embodiments can be used in any combination.

[0072] Reference throughout this specification to "one embodiment" or "preferred embodiment" means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the presently claimed invention. Thus, appearances of the phrases "in one embodiment" or "in a preferred embodiment" in various places throughout this specification are not necessarily all referring to the same embodiment but may refer to different embodiments of the presently claimed invention. Furthermore, the features, structures or characteristics may be combined in any suitable manner, as would be apparent to a person skilled in the art from this disclosure, in one or more embodiments. Furthermore, while some embodiments described herein include some, but not other features included in other embodiments, combinations of features of different embodiments are meant to be within the scope of the subject matter, and form different embodiments, as would be understood by those in the art. For example, in the appended claims, any of the claimed embodiments can be used in any combination.

DETAILED DESCRIPTION

[0073] One aspect of the invention concerns embodiments of the plastic extension which incorporate a new design resulting in increases of comfort, wearability and appearance for the user. Embodiments of the plastic extension include but are not limited to members of corresponding lengths and widths to match small, medium and large fingernails. The construction, shape and dimensions of each member of each group of embodiments of the plastic extension deliver an improved fit, minimize "pop-off", minimize edge catch and provide resistance to cracking. The shape of an embodiment of the plastic extension covering a fingernail according to the invention has increased attractiveness due to the management of the position and height of the apex in combination with the dimensions of the cuticle and the distal ends of the embodiment.

[0074] Another aspect of the invention concerns embodiments of an applique kit which are a combination of embodiments of the plastic extension, an adhesive and an optional processing aid. The adhesive encompasses one of at least four different classes of adherent materials including a UV curable gel, a cyanoacrylate, a free radical curable composition and a pressure sensitive adhesive. The optional processing aid includes an appropriate agent to enable the adhesive to attach the plastic extension firmly to the fingernail. Exemplary processing aids directed to the kinds of adhesive include a UV LED lamp, a cyanoacrylate curing agent, a free radical initiator and driers/release covers.

[0075] A further aspect of the invention concerns embodiments of a method for fitting and attaching embodiments of the plastic extension to one or more fingernails.

The Plastic Extension

[0076] Embodiments of the plastic extension according to the invention are adapted to fit fingernails from their cuticle ends to at least their distal ends. These embodiments of the plastic extension comprise a curved plastic piece having longitudinal and latitudinal curves adapted to fit the shape of the fingernail. Embodiments of the plastic extension are appropriately sized and pleasingly shaped for fitting all fingernails.

[0077] Embodiments of the plastic extension are constructed of a polymer or copolymer of acrylate and/or urethane and/or ABS plastic, preferably acrylate and/or urethane, more preferably acrylate. The cuticle end of embodiments of the plastic extension is flexible and has a thin cross-section. The distal end of embodiments of the plastic extension is relatively inflexible and has a thicker cross-section relative to the cross-section at the cuticle end.

[0078] More specifically, the embodiments of the plastic extension comprise any known overall shape as well as at yet unknown derivatives thereof. Typical, known overall shapes include but are not limited to primary overall shapes such as round, square, coffin, almond, almondetto and stiletto. Derivatives of such primary overall shapes are also included. Any overall shape, such as those known, those being derivatives thereof and those being fanciful and unknown, will have characteristics according to the invention for providing an attractive covering for a fingernail in that they will be designed to fit a natural fingernail and are included within the invention. A common feature of almost all overall shapes will provide cuticle ends that will be shaped to fit or substantially fit the cuticles of natural fingernails although square, pointed and inverse rounded cuticle shapes would also be possible as uncommon shapes. Derivatives may have variations of side and distal end designs ranging from combinations of the primary side, edge and end designs to designs that do not track the sides, edges and tips of natural fingernails at all. Some contemplated variations may include straight thin columns, curlicue edges, hammer shapes, inverse curve shapes and shapes that appear whimsical and are very impractical. For example, a derivative of a combination of square and stiletto shapes would be a squalito shape. Another would be a stiletto having a blunted distal end. Additional derivatives would be almondetto which is a combination of almond and stiletto shapes, or an oval shape which elongates the round shape. At least some of these shapes are depicted in FIG. 6.

[0079] The choice of an embodiment of the plastic extension for fitting onto a fingernail depends upon the wearer's

selection of an overall shape and its length. These two parameters do not depend upon the natural dimensions of the natural fingernail. The third selection, i.e., width/size, depends upon the width of the natural nail. Human fingernail widths may be measured at the widest dimension running across the nail at mid-length. These natural fingernail widths generally fall into width dimensions ranging from about 6 mm to about 15 mm. Accordingly, the size of the plastic extension is chosen by the width of the natural fingernail.

[0080] The width dimensions of the fingernails of a hand are correlated and related. The correlation means that fingernail widths of the thumb, index finger, middle finger, ring finger and pinky fit a pattern according to the size of the hand.

[0081] A set of plastic extensions chosen according to the fingernail widths of the hand provides a combination of plastic extension sizes (widths) of the set. While the overall shape and the desired length are chosen by the user as mentioned above, they are tempered by the length of the natural nail. For example, if the natural nail is 12 mm measured from the cuticle to the smile line, a choice of plastic extension with a length of 10 mm is not practical. It is too short. The chosen set coordinated with the fingernail widths will provide each individual plastic extension embodiment having parameters and dimensions consistent with the parameters and dimensions according to the invention.

[0082] Embodiments of a set of plastic extensions and each individual plastic extension for a hand may include any one or more of the multitude of overall shapes available. Individual plastic extension embodiments and members of the set of individual plastic extensions are illustrated by FIGS. 1, 2, 3A, 3B, 4, 5A, 5B. Further details of the embodiments of the invention are provided by the following descriptions of these Figures.

[0083] FIG. 1 illustrates the top plane view of embodiments of the plastic extension. The top surface (100) is bounded by left and right edges (501) and (502) and by cuticle end (300) having midpoint (301) and distal end (400) having midpoint (401) which ends (300 and 400) are at the termini of the longitudinal length line (600) which is a straight line running from cuticle end midpoint (301) to distal end midpoint (401).

[0084] FIG. 2 illustrates the side plane view of embodiments of the plastic extension. The cuticle (300) and distal (400) ends provide the longitudinal end points of the longitudinal curve (601) having an ascending slope (603) and a descending slope (604) which meet at apex (602) of the longitudinal curve (601). The apex (602) is positioned vertically above the apex point (606) on the longitudinal length line (600). Apex length (607) is the distance along the longitudinal length line (600) between the cuticle midpoint (301, FIG. 1) and the apex point (606). The longitudinal length line (600) is subsumed by a presumed horizontal plane between the left (FIG. 1, 501) and right (FIG. 1, 502) edges. Apex height (605) is the length of the line from apex (602) to apex point (606). The bracket designation at apex (602) indicates the cross-section of the apex shown by the inset (610).

[0085] FIG. 3A illustrates an elevated cuticle end view of the plastic extension. Cuticle end (300) has latitudinal curve (700) running from left side (501) to right end (502) and longitudinal curve (603) running up to apex (602).

[0086] FIG. 3B illustrates the side perspective view of the plastic extension. Cuticle end (300) and distal end (400) provide the longitudinal end points of the latitudinal curve (700) running from right edge (502) to left edge (501). The longitudinal curve (603, 604) combines with the longitudinal curve (700) to provide a double curved perspective shape which has been expanded outward to form the apex (602). [0087] FIG. 4 illustrates the bottom plane view of the plastic extension. Bottom surface (200) is parallel to the top surface (100) and the cross section or thickness between the top (100) and bottom (200) surfaces varies with the smallest cross section being at the cuticle end (300), the largest cross section being at the distal end (400) and a medium cross section at the apex (602). The width (503) is measured at the widest point between the left (501) and right (502) edges as marked by points (501p, 502p).

[0088] FIG. 5A illustrates the cuticle end plane view. The cuticle height (310) is measured from the presumed horizontal plane on which the left (501) and right (502) edges rest. The left (502) and right (502) edges curve upward from the presumed horizontal plane in the vicinity of the cuticle as shown in FIG. 7, digital scan of the plastic extension model. The apex height (605) is the tallest height of the plastic extension seen from the cuticle end plane view. The cross section (320) of the top (100) and bottom (200) surfaces at the cuticle end (300) is shown.

[0089] FIG. 5B illustrates the distal end plane view. The distal height (410) is measured from the presumed horizontal plane between the left and right edges at the distal end. The distal end thickness (420) is the cross section between the top (100) and bottom (200) surfaces at the distal end.

[0090] FIG. 6 illustrates some of the primary overall shapes and several derivatives thereof which may be employed as an overall shape for embodiments of the plastic extension and members of sets of embodiments of the plastic extensions.

[0091] The parameters and dimensions of the embodiments of individual plastic extensions and members of sets of plastic extensions illustrated by these Figures are provided by the following entries a–l wherein the numbers in parentheses refer to numbers of plastic extension drawings depicted on FIGS. 1, 2, 3A, 3B, 4, 5A and 5B.

[0092] a) Individual embodiments of the plastic extension and the members of a set of embodiments of the plastic extension have a longitudinal length (600) of from about 15 mm to about 37 mm, preferably up to about 35 mm, as a straight line from the midpoint (301) of cuticle end to the midpoint (401) of the distal end and running along the presumed horizontal plane between the left edge and right edge. Individual plastic extensions and the members of a set of plastic extensions fit into longitudinal length categories of:

[0093] a) short ranging from about 15 mm to about 21 mm,

[0094] b) medium ranging from about 19 mm to about 28 mm, and

[0095] c) long ranging from about 24 mm to about 37 mm.

[0096] It would appear that the short category with a length of 19–21 mm overlaps with this same length of the medium category and the same would seem apparent for the medium and long categories. This apparent overlap is explained by coordination of width segments with the length categories. Each of the three length categories have widths

coordinated with lengths such that for each length category, the widest width coincides with the longest length and the smallest width coincides with the shortest length. This coordination is divided into segments of width ranges for the length categories shown by Table A. This coordination explains why someone fitted with a 19–21 mm length plastic extension embodiment would otherwise have an issue choosing between short and medium categories. This person with a small width finger (6–8 mm) would use a medium length category plastic extension while a person with a large finger (12–15 mm) would use a short category plastic extension. Of course, a person with a small width finger (6–8 mm) or large width finger (12–15 mm) may choose to have an extremely long plastic extension such as 37 mm or longer. While impractical, such a choice would be within the long category.

TABLE A

Avg. Width (mm)	Short (mm)	Medium (mm)	Long (mm)
12-15	19-21	22-28	29-37
9.5-12	18-21	22-28	29-34
8-9.5	16-20	21-26	27-32
6-8	15-18	19-23	24-30

[0097] b) Individual embodiments of the plastic extension and the members of a set of embodiments of the plastic extension have a longitudinal curve (FIG. 1, 601) of the plastic extension running from the cuticle end to the distal end which increases in slope (FIG. 1, 603) from the cuticle end to an apex (FIG. 1, 602) and then decreases in slope (FIG. 1, 604) from the apex (602) to the distal end (FIG. 1, 400). The heights of the cuticle end, the apex and the distal end establish the slope of the longitudinal curve.

[0098] c) Individual embodiments of the plastic extension and the members of a set of embodiments of the plastic extension have an apex point (606) on the longitudinal length which is directly below and coincident with the apex (602) of the longitudinal curve wherein the length between the cuticle end of the longitudinal length and the apex point is the apex length (607). The apex length depends upon whether the longitudinal length category is short, medium or long categorized according to the length category as described by Table A. This dependency can be expressed as a ratio of the apex length to the longitudinal length and is selected from a range of from 0.27 to 0.49. The ratio of the apex length to longitudinal length provides the position of the apex along the longitudinal curve. As the total longitudinal length increases, the ratio decreases so that the apex remains in about the same position.

[0099] This variation of apex length expressed as a varying ratio of the longitudinal length presents an appearance of the plastic extension similar to the appearance of a natural nail. To accomplish this, the apex location remains in about the same location irrespective of the length of the plastic extension. This feature in part enables the user to have a “natural look” for his or her fingernail covered by the plastic extension.

[0100] The development of the “natural look” segments this ratio range by the length categories described above, Table A. The Detail Tables below provide several examples.

Table B presents a summary of these ratio variations coordinated by minimum length segments of each length category. The selection of a short, medium or long category ratio depends upon length selected as well as the width of the finger to be fitted with the plastic extension as indicated by Table A. For example, a very small width of 6 mm corresponds to lengths between 15-18 mm for a short category while this same width corresponds to lengths between 19-23 in the medium category. The ratio chosen will accordingly be in the length category corresponding to the width/length given by Table A. Ratios for sizes, lengths and any overall shape for embodiments of the plastic extensions not described by Table B or the Detail Tables can be compared and extrapolated from the Table A and B's segmented length categories and their corresponding ratios of apex length to longitudinal length or by comparison with the Detail Tables. A choice of an extremely long plastic extension such as longer than 37 mm would have an Apex to Length Ratio extrapolated from the ranges of Tables A and B.

TABLE B

Length Category	Length in mm	Apex to Length Ratio
Short	15-21	0.41-0.49
Medium	19-28	0.32-0.40
Long	24-37	0.27-0.31

[0101] When individual plastic extensions are fitted and attached to natural nails according to appropriate sizes (widths) and desired longitudinal lengths, the apex length in part enables the plastic extension to appear like a natural nail as discussed above. This variation of approximately 27% to 49% of the longitudinal length according to the inverse relationship of ratio to longitudinal length duplicates the apex and curve configurations of longer and shorter natural nails. This "natural look" is further advanced by combination of this ratio with the ratio of the cuticle to apex heights discussed below. This combination of ratios delivers a natural appearing plastic extension that does not appear artificial or unattractive as the natural nail grows. The apex position combined with the cuticle to apex height ratio are features of the embodiments of the plastic extension that provide the comfort, fit and attractive natural appearance of these embodiments attached to a finger or thumb nail plate.

[0102] d) Individual embodiments of the plastic extension and the members of a set of embodiments of the plastic extension have an apex height (FIG. 2, 605) of from about 3 mm to about 6 mm, measured vertically from the top surface at the apex to the apex point on the longitudinal length. The apex height coincides directly with the length and width of the plastic extension such that the longest length and widest width coincide with the maximum apex height of the range and the shortest length and smallest width coincide with the minimum apex height of the range.

[0103] e) Individual embodiments of the plastic extension and the members of a set of embodiments of the plastic extension have a cuticle center height (FIG. 5, 310) of from about 1.5 mm to about 3.5 mm relative to the presumed horizontal plane between the left and right edges. For each length category, the cuticle height coincides directly with the length and width of the plastic extension such that the longest length and widest width coincide with the maximum cuticle height

of the range and the shortest length and smallest width coincide with the minimum cuticle height of the range.

[0104] f) Individual embodiments of the plastic extensions and the members of a set of embodiments of the plastic extension have a ratio of cuticle height (FIG. 5A, 310) to apex height (FIG. 5A, 605; FIG. 2, 605) ranging from about 0.5 to about 0.62 for all size categories and all length categories. The ratio of cuticle to apex heights is another feature of the embodiments of the plastic extension that delivers a natural appearing plastic extension that does not appear artificial or unattractive as the natural nail grows. The ratio of cuticle to apex height combines with the ratio of apex length to longitudinal length discussed above to deliver features of the embodiments of the plastic extension that provide the comfort, fit and attractive natural appearance of these embodiments attached to a finger or thumb nail plate.

[0105] g) Individual embodiments of the plastic extension and the members of a set of embodiments of the plastic extensions have a latitudinal curve (FIG. 3a, FIG. 3b, 700) extending between the left and right edges and running from the cuticle end to the distal end.

[0106] h) Individual embodiments of the plastic extension and the members of a set of embodiments of the plastic extension have a width (FIG. 4, 503) between left and right edges of from about 6 mm to about 15 mm wherein the width is measured at the widest distance between the left and right edges.

[0107] i) Individual embodiments of the plastic extensions and the members of a set of embodiments of the plastic extension have a distal end thickness (FIG. 5B, 420) of no more than about 1.3 mm, preferably ranging from about 0.6 mm to about 0.9 mm.

[0108] j) Individual embodiments of the plastic extension and the members of a set of embodiments of the plastic extension have a cuticle end thickness (FIG. 5A, 320) of no more than about 1.0 mm, preferably from about 0.2 mm to about 0.4 mm, more preferably about 0.2 mm to about 0.3 mm, especially more preferably about 0.25 mm. The cuticle end thickness is the same for all length categories.

[0109] k) Individual embodiments of the plastic extension and the members of a set of embodiments of the plastic extension have an apex thickness (FIG. 2, XC 610) of the plastic extension of no more than about 1.5 mm, preferably no more than about 0.7 mm, more preferably from about 0.6 mm to about 0.7 mm.

[0110] l) Individual embodiments of the plastic extension and the members of a set of embodiments of the plastic extension have a distal end (e.g., tip) height of from about 0.5 mm to about 4.5 mm, preferably from about 0.5 mm to about 4 mm, more preferably from about 0.5-1 mm to about 3.5 mm. The choice of a distal end height from this range depends directly on the size (width) and the distal end shape. For example, an embodiment of a large exemplary size (width) of about 15 mm, has a tip height of 3.0 while an embodiment of a small exemplary size of about 6 mm) has a tip height of 1.5 mm.

[0111] The foregoing dimensions shapes and parameters of the individual embodiments of the plastic extension and members of sets of embodiments of the plastic extension are also depicted in FIGS. 7A-7D. FIG. 7A presents top digital

scan; FIG. 7B presents a cuticle end digital scan and FIG. 7C presents a side digital scan of a single embodiment of size 3 of the plastic extension. The length, width, apex height, apex position, and longitudinal and latitudinal curve degrees are shown in FIG. 7D which is a cross-sectional view cut at line 800 in FIG. 7A.

[0112] Commercial nail tips indicated as Commercial Natural and Commercial Sculptured have also been digitally scanned. FIGS. 8A-8D and 9A-8D present the digital scans of these commercial nail tips.

[0113] Comparison of FIGS. 8 and 9 with FIG. 7 of the embodiment of the plastic extension according to the invention illustrates the shape differences that lead to a better fit and appearance for the plastic extension embodiments according to the invention. The parameters and dimensions of the embodiments of the plastic extensions and members of sets thereof according to the invention prevent a rocking back and forth of the embodiments when placed on a flat surface and pushed at either end. The edges of embodiments of the plastic extensions and members of a set of plastic extensions are parallel and flat so that the embodiments do not rock. In contrast, the commercial nail tips rock when placed in the same arrangement. Their curved bottom edges are not flat and parallel relative such a flat surface.

[0114] The apex height, the ratio of the apex height to the cuticle end height and the ratio of the apex length to longitudinal length of individual embodiments of the plastic extension and sets thereof according to the invention deliver a longitudinal curve, an apex position and an apex location that provide comfort, fit and a natural look compared with the commercial tips (see FIGS. 8 and 9). When applied to fingernails, these heights, lengths and ratios of the embodiments of the individual plastic extension and sets thereof provide more comfortable and eye-catching appearances relative to the commercial tips.

[0115] The appearance of the individual embodiments of the plastic extensions and the commercial tips adhered to fingernails and viewed from the side or top reveal shapes dominated by the apexes of the plastic extensions and commercial tips. The parameters for the individual embodiments of the plastic extension enable appearance that is neither flat nor significantly raised relative to the natural nail plates. In contrast, the commercial tips tend to be either flat which signals unnatural flattening of the overall appearance or significantly raised at the middle which signals an undesirable thickening and discomfort. In addition, according to the invention, embodiments of the individual plastic extension and members of a set of embodiments of the plastic extension have the ratio of cuticle height to apex height, cross-sectional dimensions of the cuticle top and bottom surfaces as well as a low cuticle end Stiffness quotient (See the Flexibility shown by Table 4 below and FIGS. 10A, 10B and 10C) which easily allow the attachment at the cuticle and distal ends at same time. In contrast, commercial tip attachment must be made serially by attachment of one end and then the other because of the lack of edge flatness, the apex height and a higher cuticle end stiffness quotient (Table 4 and FIGS. 10A, 10B and 10C). The serial attachment with commercial tips provides tips that are too curved or too flat when adhered serially to the nail plates.

[0116] The dimensional and parameter features and flexibility tests further illustrating these differences are provided in the Examples section below. One example provided by FIGS. 11A, 11B and 12 shows a side by side comparison of

the plastic extension bonded with a UV cured gel to the right ring finger of an experimental test subject and a competitor plastic applique bonded with the UV cured gel to the middle finger of the same test subject. The depicted side-by-side comparison shows the differences between the plastic extension embodiment of the invention and the competitor plastic applique. In addition, the Examples section provides a testimonial description by the test subject of the comfort and fit of the plastic extension embodiment of the invention relative to that of the competitor plastic applique.

The Adhesive

The adhesive that will enable strong connection of the plastic extension embodiments and the fingernails may be any of four classes including:

- [0117] a) The UV curable gel composition;
- [0118] b) The cyano(meth)acrylate;
- [0119] c) The free-radical curable composition
- [0120] d) The pressure sensitive adhesive including but not limited to dual sided tape with PSA.

UV Gel Composition

[0121] Embodiments of the UV gel preferably enable a formation of a partially cured gel that delivers an adherent surface and a firm but flexible gel content so that placement of the plastic extension onto the gel coated nail plate and its positional adjustment to fit the plastic extension appropriately with respect to the nail plate, the cuticle and the lateral and proximal nail folds is easily accomplished. The UV gel of the preferable embodiment may be fully cured after placement of the plastic extension following the optional partial cure and positional adjustment. Alternatively, the UV gel as a non-cured embodiment can provide an adherent surface that will hold the plastic extension after its placement and positioning. Accordingly, this UV gel embodiment may be fully cured immediately after placement of the plastic extension. The UV reactive olefins of the UV gel include aliphatic and aromatic (meth)acrylate, optional hydroxyalkyl (meth)acrylates as well as a cross-linking (meth)acrylate dimer and/or trimer.

[0122] The UV curable gel embodiments deliver a UV curable process that preferably avoids unacceptable temperature rise during the polymerization process such as a temperature exceeding 58° C. Preferable embodiments enable maintenance of a partial cure state over a period of at least a few minutes to enable ready placement and moderate adhesion combined with post-placement movement when the plastic extension is fitted onto a nail plate coated with the UV curable gel.

Monomer and Dimer

[0123] The monomer compound of the UV gel may comprise a C1-C10 alkyl (meth)acrylate ester and/or a C5-C12 cycloalkyl (meth)acrylate ester and/or arylalkyl (meth)acrylate ester or a corresponding crotonate ester monomer. A dimer/trimer (meth)acrylate may be a (meth)acrylate or crotonate dimer/trimer formed of a polymeric or copolymeric link of one or more functional units of urethane, ester, amide, polyol or dimethyl siloxane between the terminal (meth)acrylate or crotonate groups.

[0124] Preferably, the monomer may be methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, t-butyl (meth)acrylate, bornyl (meth)acrylate, norbornyl (meth)acrylate,

isobornyl (meth)acrylate, cyclohexyl (meth)acrylate, benzyl (meth)acrylate, more preferably methyl and/or ethyl (meth)acrylate and/or norbornyl (meth)acrylate and/or cyclohexyl (meth)acrylate and/or benzyl (meth)acrylate.

[0125] When a monomer alone constitutes the polymerizable entity of the UV gel, a cycloalkyl acrylate/arylalkyl (meth)acrylate may be used alone or combined with a minor portion of a methyl or ethyl (meth)acrylate such as from about 2 wt % to about 25 wt %, preferably from about 2 wt % to about 15 wt % relative to the total weight of monomer. Either or both of the cycloalkyl (meth)acrylate and arylalkyl (meth)acrylate may form the complete polymerizable monomer content. When used together, they may be combined at molar ratios of 1:1 to 1:10; preferably 1:2 to 1:5. Preferred cycloalkyl, alkyl and arylalkyl (meth)acrylates include isobornyl acrylate, and benzyl acrylate and methyl acrylate with a molar ratio of 1:1:0.2 to 1:5:0.1 of the isobornyl acrylate to benzyl acrylate to methyl acrylate.

[0126] Preferably, the monomer and dimer/trimer together constitute the polymerizable entity of the UV gel. The (meth)acrylate dimer/trimer with the polymeric or copolymeric link of multiple units of urethane and/or ester and/or polyol or a combination thereof may be formed by esterifying (meth)acrylic or crotonic acid with the corresponding polymeric or copolymeric link coupler carrying terminal hydroxyl groups or by reacting a hydroxyalkyl (meth)acrylate or crotonate with a polymeric or copolymeric link coupler carrying terminal isocyanate or carboxylic acid groups. If the multiple units are alkylenyl urethane or polyol urethane, the coupler may be a di-hydroxy terminated polymeric alkylenyl or polyol urethane or a di-isocyanate terminated polymeric alkylenyl or polyol urethane, hereinafter an α,ω -dihydroxy or α,ω -di-isocyanate multiunit alkylenyl or polyol urethane coupler. If the corresponding oligomeric link is a multiunit alkenyl ester, the coupler may be a di-hydroxy terminated or a di-carboxylic acid terminated multiunit alkylenyl ester, hereinafter an α,ω -dihydroxy or α,ω -di-carboxylic acid multiunit alkylenyl ester coupler.

[0127] The dimer with a link of a multiunit amide, polyol or alkylenyl may be formed by combination of a (meth)acrylic acid or crotonic acid with the corresponding α,ω -di-amine terminal or α,ω -dihydroxy terminated oligomeric polyamide or polyol coupler through an amidation or esterification.

[0128] Preferred examples include but are not limited to a coupler at least di-terminated by a (meth)acrylate group wherein the coupler is:

[0129] a) a 1-10 unit ethylene oxide or propylene oxide oligomer, such as but not limited to polyethylene glycol 5 (i.e., polyethylene oxide 5) or glycol (ethylene oxide), or

[0130] b) trimethylolpropane (tri-terminated), or

[0131] c) an α,ω -di-isocyanate 2-22 unit C3-C9 alkylenyl urethane or a (poly C2-C5 alkylenyloxy) urethane copolymer polymer such as but not limited to a polyurethane formed from a C5 and C6 alkane diol and isophorone diisocyanate or poly (1,4-butane diol)22— isophorone diisocyanate copolymer,

[0132] d) a polyurethane having two or more terminal (meth)acrylate groups such as but not limited to polyurethane 57, or

[0133] e) a combination of two different exemplary dimers.

[0134] The monomer and dimer concentrations may be arranged so as to provide moderate to high cross linking. A higher concentration of the dimer relative to the monomer will deliver higher crosslinking and result in a stronger binder film. The total concentration of monomer and dimer together may range from about 60 wt % to about 99 wt percent relative to the total weight of the UV gel composition, preferably from about 70 wt % to about 98 wt %, more preferably from about 75 wt % to about 95 wt %. The molar ratio of monomer to dimer wherein a mol of dimer with two unsaturations is counted as one mol may range from about 20:1 to about 2:1, preferably about 10:1 to about 4:1, more preferably about 10:1 to about 5:1.

[0135] The UV gel may typically be formed without additional non-reactive solvent. However, at least some additives may be appropriately combined with the UV gel monomer and dimer through use of an inert organic solvent including but not limited to one or more C1-C6 monoalcohols, C2-C6 diols, C3-C6 ketones, C1-C3 alkyl acetates, chloroform, ethylene carbonate, propylene carbonate, C5-C10 hydrocarbons and any mixture thereof. Preferably, the organic solvent may be methanol, ethanol, propanol, diacetone alcohol, acetone, methyl ethyl ketone, methyl acetate, ethyl acetate, butyl acetate chloroform, ethylene carbonate, propylene carbonate, hexane, heptane, octane and any combination thereof. The concentration of organic solvent optionally present in the composition may range from about 0.5 wt % to about 15 wt %, preferably from about 0.5 wt % to about 10 wt % relative to the total weight of the UV gel composition.

[0136] A photoinitiator such as benzophenone, phosphine oxide, TPO, TPO-L, 1-hydroxycyclohexyl phenyl ketone, and similar type I and type II photoinitiators may be combined with the UV monomer/dimer gel to enable photopolymerization. Typical photoinitiators include but are not limited to 2-hydroxy-4'-(2-hydroxyethoxy)-2-methylpropiophenone, 1-hydroxycyclohexyl phenol ketone, bis(diethylamino)benzophenone and the phosphinoyl and phosphinates such as ethyl(2,4,6-trimethylbenzoyl)phenyl phosphinate, phenyl bis(2,4,6-trimethylbenzoyl phosphine oxide and 2,4,6-trimethylbenzoyl diphenyl phosphine oxide. The concentration of photoinitiator may range from 0.01 wt % to about 2 wt %, preferably 0.01 wt % to about 1.5 wt % relative to the total weight of the UV gel composition.

[0137] Preferably, two photoinitiators may be included. The first one may have an absorption maximum at about 320 to 360 nm and the second one may have an absorption maximum at about 390 to 400 nm. The first photoinitiator having the shorter wavelength will initiate polymerization at the UV gel surface but is not efficient for such initiation at a gel depth. Furthermore, the first photoinitiator is inefficient at penetrating the plastic extension to perform a final cure of the UV gel when the plastic extension is in place on a fingernail with the UV gel between acting as an adherent material. The second photoinitiator having the longer wavelength is more efficient at penetrating to the depth of the UV gel and passing through the plastic extension to perform a final UV gel cure. This use of two photoinitiators so as to provide a preferential partial cure of the UV gel enables an easy fit of the plastic extension onto the fingernail.

[0138] Selective excitation of the two photoinitiators may be accomplished by use of the UV lamp having LEDs designed to produce corresponding narrow UV maximum radiation wavelengths. The selection of the shorter wave-

length LED optionally coupled with a short radiation period will at least partially cure the surface of the UV gel. The partial cure will provide a tackiness to the UV gel surface so that fitting the plastic extension onto a fingernail having thereon such a partially surface cured UV gel can be readily accomplished. The tackiness will provide an adherence to the plastic extension on the fingernail while the semifluid internal gel consistency will enable easy adjustment of the fit to the fingernail. Once the plastic extension is appropriately fitted on the fingernail, the longer wavelength UV LED may be actuated along with the shorter wavelength UV LED to complete the curing of the UV gel. This preferred method of fitting of the plastic extension to a fingernail enables an efficient placement of the plastic extension with little or no difficult “do-over.”

[0139] The UV gel may also optionally include one or more ancillary agents to provide film forming pre-formed polymers, rheology control, solubilization of individual components that are insoluble in each other, free radical scavenging, oxygen inhibition and scavenging,

[0140] A preformed film former (PFF) may optionally be included with the UV gel as an ancillary agent. The PFF may also enable UV gel pre-setting so that photopolymerization of the UV gel and especially the UV gel composed entirely of monomer may be facilely accomplished. Additionally, and optionally, a PFF having physical characteristics making it form a two phase solid or PFF domains with a continuous phase of the cured UV gel will enable at least in part an easy, mild removal by organic solvents such as acetone, methyl ethyl ketone and similar solvents.

[0141] The PFF comprises a preformed non-reactive film forming polymer such as nitrocellulose, cellulose acetate, cellulose acetate butyrate (CAB), tosylamide epoxy resin, acrylates copolymer, polyacrylates such as but not limited to PMA (polymethacrylate) and PMMA (polymethylmethacrylate), a polyester such as adipic acid/neopentylglycol/trimellitic anhydride copolymer and/or a styrene/acrylates copolymer and any mixture thereof. Preferably, the PFF is one or more of CAB, an acrylates copolymer, tosylamid epoxy resin, PMA and/or PMMA, more preferably CAB, polyester, PMA or PMMA, especially more preferably CAB or PMMA. The concentration of the PFF in the UV gel may range from about 1 wt % to about 10 wt %, preferably from about 1 wt % to about 5 wt % relative to the total weight of the UV gel.

The UV Lamp

[0142] The UV lamp may be any that is capable of producing UV radiation of the UVA and UVB regions that would interact with the photoinitiator or photoinitiators at least in part according to the UV wavelength maxima of the photoinitiators. The UV lamp may be a mercury vapor lamp or the more recently developed LED lamp. The mercury vapor lamp produces a broad spectrum of UV wavelengths while the LED lamp produces a UV spectrum that usually has a single wavelength maximum and a narrow bandwidth. Because of its more accurate control of the UV wavelength spectrum, the LED UV lamp is preferred.

[0143] A preferred UV lamp is an LED lamp with one or more LED semiconductors producing UV maxima in the range of 300 to 420 nm. More preferably, the UV lamp includes a group of LED semiconductors producing UV maxima in this range. Especially more preferably the UV lamp includes at least two groups of LED semiconductors in

which a first group produces UV maximum in the range of about 300 to about 360 nm, preferably about 320 to about 360 nm and typically a maximum of about 340 to about 360 nm. The second group of these LED semiconductors produces a UV maximum in the range of about 360 to about 420 nm, preferably about 370 to 420 nm, and typically a maximum of about 380 to about 420 nm.

[0144] The design of the UV lamp may be any that enables UV light to be directed to one or more fingernails. The design may take the form of a “C” shape with the LED semiconductors attached to the inside of the upper part of the “C”, the lower part configured to be a platform base for supporting the UV lamp on a flat surface and the middle part being the space into which the fingernail or fingernails are placed for UV radiation. Controls for managing the LED semiconductors may be positioned on the back of the “C” shape and controlling electronic circuit placed into the base.

[0145] An alternate and preferred design is directed to a bell-shaped hood with lamp sockets into which the LED semiconductors are to be placed, a flexible neck attached to the bell-shaped hood and a base-stand connected to the flexible neck for supporting this “goose-neck” lamp on a flat surface. The base stand may contain the controls for managing the LED semiconductors. The controls may include on-off switches, intensity dials and timing dials which are connected to an electronic control integrated circuit board in the base stand.

[0146] Embodiments of the UV lamp incorporate software control for management of exposure time, UV light intensity and individual variation of the intensities of the UV maxima associated with the photo initiator UV excitations at the ranges having maxima at 360 and 430 microns.

[0147] These designs for UV lamps may include software programed into the integrated circuit board for management of the LED semiconductor on-time, intensity, and which UV range of LED semiconductor is actuated. The software designed in this manner enables a first selective UV radiation at about 300 to about 360 nm and a second selective UV radiation at about 360 to about 420 nm. Preferably the selective UV radiation spectra are a first spectrum at about 340 to about 360 nm and a second spectrum at about 380 to about 420 nm. Time and intensity controllers to manage independently each of the first and second spectra may also be incorporated into the software.

The Cyano(Meth)Acrylate Composition

[0148] The cyano(meth)acrylate class comprises any one or more of the C2-C10 alkyl cyano(meth)acrylate esters including the ethyl, propyl, butyl, pentyl, hexyl, heptyl, octyl, nonyl and/or decyl esters. A well-known cyanoacrylate is methyl cyanoacrylate ester trademarked as “SuperGlue.” This “SuperGlue” kind of cyanoacrylate, methyl cyanoacrylate, will rapidly polymerize in the presence of air because of the residual moisture in air. Although this kind of methyl cyanoacrylate ester may be formulated as the primary component of the adhesive, its polymerization in contact with air is so rapid that its use on a nail plate for fitting and adhering a plastic extension have a hurried pace for completion. Consequently, it is preferred to apply the higher alkyl cyano(meth)acrylates so that the rate of the polymerization in air is slow enough to enable fitting at a reasonable pace.

[0149] Accordingly, the alkyl cyano(meth)acrylate including the ethyl, butyl, hexyl, octyl and decyl esters are preferred for this class of adhesive. The rate of polymeriza-

tion may also be controlled by use of the cyanomethacrylate ester instead of the cyanoacrylate ester. The methyl group substitution of the cyanoacrylate slows the corresponding rate of the polymerization.

[0150] Preferably, application of this class of adhesive may be performed with ethyl, propyl, butyl, hexyl and/or octyl cyanoacrylate and/or a mixture thereof, more preferably the ethyl, butyl and/or octyl cyanoacrylate and/or a mixture thereof, and especially more preferably the ethyl and/or octyl cyanoacrylate and/or a mixture thereof. A mixture of the lower alkyl cyanoacrylate with alkyl cyanomethacrylates and/or higher alkyl cyanoacrylates may be appropriate in circumstances where some degree of rapid polymerization is desired to enable a partial cure so that the plastic extension may be fitted and adjusted on the fingernail followed by a slower rate of polymerization to adhere firmly the plastic extension to the fingernail. Such mixtures may include a minor weight percentage of a lower alkyl cyanoacrylate ester having fast polymerization rate such as a methyl and/or ethyl cyanoacrylate and a major weight percentage of a cyanomethacrylate ester and/or higher alkyl cyanoacrylate having a slower polymerization rate such as a butyl or octyl cyanoacrylate or cyanomethacrylate. Weight percentages for the fast polymerization rate cyanoacrylate ester may range from about 1 wt % to about 20 wt %, preferably from 1 wt % to about 25 wt %, relative to the total weight of the composition with the remainder weight percentage being a slower polymerization rate cyanomethacrylate ester and/or higher alkyl cyanoacrylate. The total weight percentage of all cyano(meth)acrylate esters (i.e. all of the cyanoacrylate and cyanomethacrylate esters) used for the primary component of this class may range from 90 wt % to 99 wt %, preferably 96 wt % to 99 wt %.

[0151] The cyano(meth)acrylate composition may also optionally include an anhydrous, non-hydroxyl, volatile organic solvent. Preferably, the concentration of the solvent is from about 0.5 wt % to about 10 wt %, more preferably from about 0.5 wt % to about 5 wt % with the remainder being the cyano(meth)acrylate ester itself. Examples of solvent for this class of adherent liquid includes ethylene carbonate, propylene carbonate, chloroform, carbon tetrachloride, and/or a hydrocarbon such as hexane, heptane, octane. While esters and ketones such as ethyl acetate and/or methyl ethyl ketone may be incorporated as solvents for this class, they preferably are conditioned to be anhydrous and are combined with the cyano(meth)acrylate ester under very dry conditions.

[0152] When the unit container with the cyano(meth)acrylate is opened, moisture from the atmosphere will be introduced and will initiate polymerization. With a choice of cyano(meth)acrylate that has a less than extreme polymerization rate, this phenomenon will generate adhesiveness and/or increased adhesiveness for the cyano(meth)acrylate while enabling a reasonable fitting of the plastic extension.

[0153] A processing aid for the cyano(meth)acrylate adhesive may not be necessary in view of the moisture in the air. However, when needed, water may be delivered as a constituent of an organic solvent and optionally with an amine accelerant diluted with an organic solvent. The amine accelerant may be a tertiary amine such as alkyl diphenyl amine, triphenyl amine or methyl dihexyl amine. The concentration of the amine accelerant and water may be in a range of from 0.1 wt % to 2 wt %, preferably 0.2 wt % to 1.5 wt % in organic solvent and the organic solvent concentration may

be in a range of from about 2 wt % to about 10 wt % relative to the total weight of the adhesive. The usual method for carrying out the application of the cyanoacrylate adhesive calls for application of the cyanoacrylate without any processing aid and is accomplished by brushing, spraying, dripping, flooding or otherwise contacting and the fingernail with cyano(meth)acrylate.

The Free Radical Curable Composition

[0154] The (meth)acrylate free radical composition comprises two separate components that are combined for use. The first component is one or more C1-C8 alkyl (meth)acrylate esters, an amine accelerant, and optional (meth)acrylate dimer, an optional PFF, an optional inert, volatile organic solvent and an optional free radical scavenger. The second compound is the peroxide initiator with optional inert bulking material.

[0155] Preferably, the alkyl (meth)acrylate may be a methyl, ethyl propyl, butyl, pentyl or hexyl (meth)acrylate with the acrylates preferred relative to the methacrylates. More preferably, the alkyl(meth)acrylate may be a methyl, ethyl or propyl (meth)acrylate with the acrylates being preferred relative to the methacrylates. The (meth)acrylate composition may also include a higher carbon count alkyl or cycloalkyl (meth)acrylate such as hexyl, cyclohexyl, decyl, isobornyl or norbornyl (meth)acrylate. The presence of such a higher carbon count alkyl (meth)acrylate may act as a reactive diluent to facilitate continued polymerization after the volatile organic solvent has dissipated. Mixtures of C1-C3 alkyl (meth)acrylate and C3-C8 alkyl (meth)acrylate with optional cycloalkyl (meth)acrylate may also constitute the (meth)acrylate monomer mixture. The lower alkyl (meth)acrylates may exhibit higher polymerization rates than the higher alkyl and cycloalkyl (meth)acrylates so that the adhesive may rapidly be at least partially cured to enable fitting in place of the plastic extension. The concentration of the (meth)acrylate monomer as the primary component of the adhesive may range from about 70 wt % to about 99 wt %, preferably from about 80 wt % to about 99 wt % relative to the total weight of the adhesive.

[0156] Optionally, the free radical curable composition may include a (meth)acrylate dimer or trimer cross linker formed of an oligomeric link of one or more, preferably one to six, more preferably one to three linear units of urethane or ester with terminal (meth)acrylate groups or a single branch oligomeric trimeric link of a three or four urethane or ester units with three terminal (meth)acrylate groups. These dimers and trimers may be selected from the dimers and trimers described above for the UV curable gel.

[0157] Optionally the free radical monomer composition may include a PFF as described above for the UV curable gel. The concentration of the PFF in the free radical curable monomer composition may be the same as given for the UV curable gel.

[0158] Polymerization initiation for the (meth)acrylate is generated by the second component, the processing aid, which is a peroxide. Preferably, the second component is a solid peroxide such as dibenzoyl peroxide or dicumyl peroxide, although a liquid such as hydrogen peroxide may also be employed. While the peroxide is to be combined with the (meth)acrylate monomer and optional amine accelerant to initiate polymerization to initiate free radical polymerization, it cannot be stored with the latter two components. Consequently, the peroxide may be maintained in a separate

container as the second component. To initiate polymerization with the (meth)acrylate monomer, a portion of the peroxide may be mixed with the (meth)acrylate composition (adhesive) immediately before use. The concentration of peroxide to be used with the (meth)acrylate class of adhesive liquid may range from about 0.1 wt % to about 3 wt %, preferably from 0.1 wt % to about 2 wt %, more preferably about 0.1 wt % to about 1 wt % relative to the total weight of the composition containing the adherent liquid to be applied to the nail plate. The concentration of peroxide in optional inert volatile organic solvent may be back calculated to provide the foregoing concentration of peroxide.

[0159] The polymerization of the (meth)acrylate monomer can be accelerated by the secondary or tertiary di or tri-C3-C6-alkyl or di or tri-aryl or alkaryl amine, preferably a tertiary amine once polymerization is initiated by the peroxide. The amine accelerator may be combined with the (meth)acrylate monomer at any time before use and preferably is combined with the (meth)acrylate monomer in the unit container of the kit described below. Optionally, when the amine accelerator and (meth)acrylate monomer are provided as a mixture in a single container for the kit, an appropriate amount of a free radical scavenger may be included to prevent premature free radical polymerization of the monomer. The concentration of the amine accelerator may range from about 0.1 wt % to about 5 wt %, preferably from about 0.1 wt % to about 3 wt %.

[0160] The (meth)acrylate adhesive may optionally include an inert organic solvent including but not limited to one or more C1-C6 monoalcohols, C2-C6 diols, C3-C6 ketones, C1-C3 alkyl acetates, chloroform, ethylene carbonate, propylene carbonate, C5-C10 hydrocarbons and any mixture thereof. Preferably, the organic solvent may be methanol, ethanol, propanol, diacetone alcohol, acetone, methyl ethyl ketone, methyl acetate, ethyl acetate, chloroform, ethylene carbonate, propylene carbonate, hexane heptane, octane and any combination thereof. The concentration of organic solvent optionally present in the (meth)acrylate class of adherent liquid may range from about 0.5 wt % to about 30 wt %, preferably from about 0.5 wt % to about 20 wt % relative to the total weight of the adherent liquid.

[0161] Typically, the free radical curable composition is packaged in two units, one for the monomers, optional dimer/trimers, optional PFF and accelerator and another for the free radical initiator such as benzoyl peroxide. Commonly this two unit preparation is termed a liquid-powder composition wherein the powder is the solid benzoyl peroxide.

The Pressure Sensitive Adhesive

[0162] The pressure-sensitive adhesive (PSA) may be based on block copolymers composed of olefinic copolymers or olefinic aromatic copolymers or acrylic esters or rubber polymers or silicone polymers. Typically, one of the blocks is an elastomeric block while the other block is stiff or hard. The combination of elastomeric and hard blocks provides the adherence from the elastomeric blocks and resistance to separation from the hard blocks. Alternatively, the PSA may be a homogeneous mixture of one or more elastomeric polymers and one or more rigid or structural polymer.

[0163] The olefinic/olefinic aromatic PSA may be composed of polymer blocks of vinyl aromatic compounds (A blocks), preferably styrene, and those formed predominantly

by polymerization of 1,3-dienes (B blocks), such as, for example, butadiene and isoprene, or a copolymer of the two. The PSA's can also be partially or completely hydrogenated in the diene block. Block copolymers of vinyl aromatic compounds and isobutylene may also be used. Instead of the polystyrene blocks, vinyl aromatic compounds and other homo- and copolymers comprising aromatic compounds (preferably C₈- to C₁₂-aromatic compounds) having glass transition temperatures of greater than 75° C., such as, for example, aromatic blocks comprising α -methyl styrene may be used.

[0164] The pressure sensitive adhesive may alternatively be composed of an AB block copolymer of acrylate ester and vinyl acetate monomers. Each A block, which has a T_g of at least 50° C., i.e., the hard block, may be the reaction product of a first monomer composition that contains an alkyl methacrylate, an aralkyl methacrylate, an aryl methacrylate, or a combination thereof. The B block, which has a T_g no greater than 20° C., i.e., the soft elastomeric block, is the reaction product of a second monomer composition that contains an alkyl(meth)acrylate, a heteroalkyl(meth)acrylate, a vinyl ester such as vinyl acetate, or a combination thereof. The block copolymer contains 20 to 50 weight percent A block and 50 to 80 weight percent B block based on the weight of the block copolymer.

[0165] The pressure sensitive adhesive (PSA) useful according to the invention may be a commercially available water-based or solvent-based acrylic adhesive, a natural or synthetic rubber adhesive, a styrene block copolymer, a silicone adhesive or other similar organic polymer capable of forming mechanical, polar and hydrogen bonding interactions with a solid substance.

[0166] Exemplary PSA's include those from 3M, Henkel, Arkema, Ellsworth Adhesives, H. B. Fuller, Avery Dennison and others. A typical PSA is an acrylate PSA formed from a combination of acrylic acid and acrylate esters such as ethylhexyl acrylate, isooctyl acrylate, butyl acrylate and methyl acrylate. Examples include Crilat, Aberdingk A C, Arkiril, Acronal, Eastman Ex, Glocryl, Acryltac and others.

[0167] The PSA may be applied directly to a portion of the bottom surface of the plastic extension that would cover the natural fingernail. The exposed surface of the PSA may be covered with a release film comprising a silicone treated paper or plastic film to protect the PSA from unwanted adherence. The release film may be removed from the PSA coated plastic extension shortly before application of the PSA coated plastic extension to the natural nail.

[0168] Alternatively, a two-sided adhesive film having coatings of PSA on both sides and having release papers covering both coatings may be used according to the invention. Preferably, the film side to contact the plastic extension is coated with a PSA that develops a stronger bonding arrangement than the PSA in the film side to contact the nail surface. With this arrangement, removal preferentially breaks the PSA bonds on the nail side so as to avoid and/or minimize damage to the nail plate. Typically the two-sided adhesive film is provided as a pad having a shape and size fitting the bottom surface portion of the plastic extension that will contact the nail plate. A group of pads may be provided with the applique kit, described below, as separate parts. In use, the release paper on the side to contact the plastic extension is removed and the pad attached to the bottom surface of the plastic extension. The release paper on

the side to contact the nail plate is then removed and the plastic extension with PSA pad is fitted to the nail plate.

The Nail Applique Kit

[0169] The Kit embodiments according to the invention comprise single units of the adhesive, applicator(s) and a selection of embodiments of the plastic extension sized as discussed above and below. The preferred nail applique kit includes single units of the UV gel and applicator(s) along with selections of embodiments of the plastic extension. The units of UV gel may be combined with a cap optionally holding a brush for application of the composition to one or more nail plates. The Kit embodiments according to the invention also comprise a selection of embodiments of the plastic extension as sets of appropriate sizes relative to the natural fingernail widths of the person to whom the sets are to be applied. An assortment of emery nail files, scissors and “orange stick” shapers for trimming the plastic extensions after they are bonded to nail plates.

[0170] The Kit may be packaged in a box or other appropriate packaging construct for holding the container unit or units, brushes and other materials needed for practice of the method according to the invention. Included in the Kit box may also be written instructions for application of the UV gel to nail plates, fitting the plastic extensions to the nail plates and stepwise UV curing the UV gel coating on the nail plates.

[0171] The Kit further includes the UV lamp described above and designed for step-wise curing one or several nails coated with UV gel and plastic extensions.

[0172] Similar Kits for the cyanoacrylate, free radical curable composition and PSA adhesive may be constructed according to the foregoing description with appropriate changes for the adhesive class involved.

The Method

[0173] Embodiments of the method incorporate either two kit components or three kit components depending on whether a processing aid is needed. For cyanoacrylate, free radical curable composition and PSA's, a two component kit is typical with a third optional component being the processing aid. The two component kit may include a selection of embodiments of the plastic extension for fitting different sized fingernails and one or more unit containers of the adhesive along with optional processing aid. The processing aid for the cyanoacrylate class may be an organic solvent with water and optional amine accelerator. The processing aid for the free radical addition composition may be the solid peroxide and a separate unit of solvent. For the UV gel, a three component kit is provided including the adhesive UV gel, embodiments of the plastic extension, and the UV lamp. The unit container of UV gel includes application apparatus, such as a brush or sponge for providing application of the UV curable gel to the nail plate.

[0174] Attachment of the selected embodiment of the plastic extension to a natural nail according to the invention provides an extended nail configuration that is attractive, long lasting, comfortable, provides a shape like a natural nail and avoids edge incidents that lead to pop-off. The attachment may be practiced with any of the above-described adhesives and is preferably practiced with the UV curable gel.

[0175] Embodiments of the method according to the invention are directed to adhering one or more embodiments of the plastic extension to one or more fingernails with the adhesive. The method can prepare one finger at a time or multiple fingers or whole hands can be prepared by carrying out each part of the sequence of steps on all of the multiple fingers or the whole hand of fingers. As a preliminary to the first step, choice of appropriate sized embodiments of the plastic extension from a selection of many sizes may be made by dry fitting a selected embodiment to the desired nail plate. Once an appropriate size of the embodiments of the plastic extension has been selected, the steps for bonding the embodiment to the nail plate through use of the adhesive may be practiced.

[0176] Preferably, the method is practiced with the UV gel coated on one or more fingernails and curing the UV gel with the UV lamp. The method preferably utilizes a double UV radiation involving at least partially or flash irradiating the UV gel on the fingernail before the embodiment of the plastic extension is placed onto the nail and again completing the cure of the UV gel with UV radiation after than embodiment is fitted onto place on the fingernail.

[0177] In the first step of this preferred method, the ready to use UV gel is applied from its unit container by forming a coating of UV gel on the nail plate. One pass or more passes of the applicator carrying the UV gel delivers an even coat of the UV gel to the nail plate. The UV gel preferably is thixotropic so that during application, it will flow to produce a substantially to essentially even coating on the nail plate. Because of its thixotropic property, the UV gel will not flow in its quiet state after application so that the coating will remain as applied.

[0178] The embodiment of the plastic extension may be fitted onto the nail plate coated with UV gel without further manipulation and/or management of the UV gel. The UV gel preferably will have at least some adherence without further UV gel management to hold the embodiment in place after fitting.

[0179] Alternatively, the UV gel coating may be flash irradiated to provide a tacky state of the flash cured UV gel. Typically, the flash irradiation tackifies the surface of the UV gel but does not crosslink and/or otherwise cure the entire UV gel coating. The flash irradiation to provide surface tackiness may be accomplished with LED semiconductors providing a full UVA and UVB range of radiation over a brief period of time such as 5 to 10 seconds. Alternatively, the UV gel may be irradiated with LED semiconductors producing UVB radiation in the range of about 300 to about 360 nm, preferably in the range of about 340 to 360 nm for a time sufficient to render the surface of the UV gel tacky. The tacky UV gel surface delivers an adherence to the plastic extension to hold it in place when fitted onto the nail plate. The interior of the UV gel at this step remains at least substantially to essentially uncured.

[0180] The embodiment of the plastic extension is pressed or otherwise managed to produce essentially complete contact between the surfaces of the plastic extension and the natural nail. Complete contact is maintained by the tacky character of the partially irradiated UV curable gel. The embodiment may be re-positioned, if necessary, at this point because the UV curable gel is only tacky, not fully cured.

[0181] Because the cuticle end of the embodiment of the plastic extension is highly flexible owing to its combination of parameters described above, the plastic extension flexes

during the management of complete contact of the surfaces of the artificial nail extension and the nail whereby substantially to almost completely, all void space between these two surfaces is eliminated, except for the residual void space at the apex of the bottom surface of the plastic extension. The flexing or maintenance of complete contact preferably causes the adhesive composition to flow into this residual void space so as to almost to completely fill the void space at the apex. In this manner, the bottom surface of the embodiment of the plastic extension at the apex region also becomes bound to the natural nail surface.

[0182] In the tacky state produced by flash or UVB only irradiation, the UV gel is thixotropically fluid enough so that the embodiment of the plastic extension may be readily repositioned on the nail plate without extreme disruption of the UV gel coating. Any excess UV gel at the edges of the nail plate may be easily removed without disruption of the plastic extension fit on the nail plate. Once the embodiment of the plastic extension is appropriately fitted and positioned on the nail plate, the resulting construction of nail plate, the tacky UV gel coating and plastic extension may be UV irradiated using the LED semiconductors providing full UVA and UVB radiation to thoroughly cure the UV gel and affix the plastic extension in place. This step forms bonding between the natural nail surface and the bottom surface of the embodiment.

[0183] Additionally, the practice of method employing the intermediate step for producing the tacky surface of the UV gel enables preparation of multiple fingernails at the same time. Each of the several embodiments of the plastic extension may be easily fitted and held in place by this method. The UV cure does not continue after the UV radiation is stopped so that the tacky surface remains to hold the embodiment of the plastic extension in place but it can be repositioned at leisure over a reasonable period of time. The ability to leisurely reposition enables facile preparation of multiple plastic preparations by practicing each step on multiple nail plates before moving to the next step.

[0184] The use of acrylic, urethane or ABS plastic, preferably acrylic plastic, for embodiments of the plastic extension, its combination with the UV gel and method characterized above enables a fully cured UV coating that preferably bonds to a greater extent to the nail plate than to the plastic, preferably acrylic plastic, of the embodiment of the plastic extension. This preferred property enables removal of the embodiment from the nail plate without also removing a layer or microlayer of plate tissue at the same time. The cured coating preferably strongly bonds to the nail plate and protects it from damage during the removal of the embodiment. The cured coating in this instance may be removed by known techniques available for removing UV cured nail polish.

Formation of the Artificial Nail Extension

[0185] The plastic for forming embodiments of the plastic extension according to the invention may be any commercial thermoplastic acrylic, urethane, or ABS plastic, preferably acrylic or urethane, more preferably acrylic thermoplastic suitable for use in forming nail tips and in particular, embodiments of the plastic extension according to the invention.

[0186] The plastic material may be used in the form of pellets, or other comminuted materials to begin the process of forming the one or more plastic extensions by an injection

molding method or a 3-D printing method according to the invention. In either method, the pellets or comminuted materials are melted and extruded into an injection dye casting or to a 3-D printing nozzle.

[0187] The injection molding method utilizes an individual casting die having at least two separable parts which together form an internal cavity conforming to the size and shape of the desired individual embodiments of the plastic extension. The plastic pellets may be dried in a dryer with heat before beginning the dye casting. The dried pellets are then transferred to the feed zone of an extruder with temperature ranges suitable for beginning the melt extrusion of the chosen plastic material. From the feed zone, the heated pellets enter the extrusion zones with temperatures sufficient to form molten plastic material. The die/mold temperature is set to approximately a similar range of temperature for casting the individual plastic extensions. Following casting, the molten plastic in the individual dies is allowed to cool. The two parts of the casting dies are separated and the solidified individual embodiments of the plastic extension removed and optionally buffed to remove injection sprue. This process produces as volumetric castings embodiments of the plastic extension according to the invention.

[0188] A series of casting dies for the injection dye cast method may be created by Computerized Numerical Control (CNC) machining, or electrical discharge machining (EDM), or 3-D printing using a substrate such as silicone rubber or metal. The casting dies are configured to produce embodiments of the plastic extension having the parameters and parameter ratios described above.

[0189] The 3-D printing method can also be utilized to produce the series of embodiments of the plastic extension directly. This method utilizes an industrial 3-D printing machine, associated industrial 3-D software and a prototype computer model of embodiments of the plastic extension. Sectioned digital images of the prototype are entered into the 3-D software, and program parameters for all desired sizes and shapes of embodiments of the plastic extension are entered according to 3-D software instructions. Molten plastic material is delivered to the 3-D printing machine which serially will produce desired groups of shapes and sizes of embodiments of the plastic extension.

EXAMPLES

[0190] A size 3 embodiment of the plastic extension having a coffin overall shape and parameters following the above-described parameters and ratios was prepared by a hand shaping method using an acrylic plastic sheet and shaping the sheet with shaving and shaping tools and heat. When the shape and dimension parameters met the foregoing specifications, the model embodiment of the plastic extension was duplicated in sizes ranging in width sizes from DO to 9 (mm widths from 6 to 15) as described below.

[0191] A digital image of the top view, the cuticle end view and the side view of the size 3 plastic extension was made and is shown in FIG. 7. The digital image was generated through the use of a CAD facilitated digital optical scanner. The digital image enables accurate measurement of plastic extension parameters and ratios.

[0192] The parameters and ratios taken from the size 3 embodiment were used to generate size DO to 9 embodiments of the plastic extension as CAD facilitated digital images. The digital images were programmed into a die

design machine and the dies used to prepare plastic extensions of sizes DO to 9 were produced.

[0193] Using the same CAD facilitated digital optical scanner, examples of commercial plastic nail tips were examined and their digital images generated. These digital images are shown in FIGS. 8 and 9. FIGS. 8 and 9 provide accurate readings of the parameters and parameter ratios for these two commercial nail tips.

[0194] Using precision micrometer measuring devices, the hand measurements of embodiments of the plastic extension of FIG. 7 and the commercial tips FIGS. 8 and 9 were made and compared with the digital image measurements. The comparisons showed that the precision micrometer measurements and the digital image measurements were the same. This comparison enabled the use of precision micrometer measurements of selections of the commercial tips of various sizes and of the selections of embodiments of the plastic extension made according to the above-described programmed production. A selection of sizes of the commercial tips and embodiments of the plastic extension of the invention ranging from small to medium to large were examined by precision micrometer measurements. These micrometer measurements are listed below in the tables.

[0195] Comparisons of the parameters of these programmed production embodiments of the plastic extension and the selection of commercial tips were made to illustrate distinct differences therebetween. These parameters and differences are produced in the following Tables 1, 2, 3. The measurements are in mm. The categories length and width are the longitudinal length and the size as width. The height is the apex height. The tip height is the distal end height. The cuticle height is self-explanatory. The apex location from cuticle end is the apex length. The tip height to total height is the distal end height to apex height. The cuticle height to total height is the cuticle height to apex height. The apex length to total length is the ratio of apex length to longitudinal length.

[0196] The Summary Chart of Tables 1, 2 and 3 provides the comparisons of the apex length to total length, cuticle height to apex height and distal end height (tip height) to apex height for the embodiments of the plastic extension according to the invention and the commercial appliques (commercial Natural and commercial Sculptured). The combination of these ratios delivers the natural appearance, comfort and wearability of the embodiments of the plastic extension of the invention.

[0197] Additionally, flex tests of the cuticle ends of the programmed production embodiments of the plastic extension and of the selection of commercial tips were made. The test was conducted with texture analyzer manufactured by Texture Technologies.

[0198] Procedure A: The texture analyzer is operated according to the following steps. The Texture analyzer machine is mechanically operated and has a pressure bar calibrated to apply force in grams per second. The test piece is placed on the anvil platform directly under the pressure bar and arranged so that the pressure bar contacts the cuticle end or the distal end of the test piece. The analyzer is actuated and the pressure bar pressed against the test piece with increasing, calibrated force until the tested portion of the test piece collapses to the anvil. The collapse is indicated digitally by the texture analyzer by a sharp change in the degree of pressure being applied. The results are provided in grams per second of force for flexing the test piece.

[0199] The procedure was conducted on several test pieces for each kind of embodiment of the plastic extension and commercial tip as follows. The distal end of the embodiment of the plastic extension or commercial tip was placed in the vise of the texture analyzer and its cuticle end positioned underneath the pressure bar. The vise was tightened against the distal end to hold the test piece in place. Stiffness is the extent to which an object resists deformation in response to an applied force. The complementary concept is flexibility or pliability: the more flexible an object is, the less stiff it is.

[0200] The pressure bar was activated and the degree of pressure required to cause full flexing collapse of the cuticle end was measured. The plot of the degree of pressure as a function of displacement applied to a typical polymer material shows a uniformly ascending slope indicating elastic deformation until the point at which elasticity of the test piece or its ability to revert to original form upon release of pressure fails and the test piece becomes plastic, i.e., permanently assumes its "stretched form" and no longer resists the applied pressure. At this point the plot shows a sharp drop-off and a rapidly descending slope.

[0201] Table 4 provides the results of the flex tests. It tabulates the Stiffness (gm/mm) of the cuticle ends of 3 different embodiments of the coffin overall shape plastic extension and compares these results with the Stiffness of corresponding tips from commercial Sculptured and commercial Natural portfolio in Coffin Medium category. These three tip sizes were chosen to have full range of large, medium and small sizes. The corresponding graphs of flex pressure over time (i.e., Stiffness) for each of these size categories of embodiments of the plastic extensions and commercial tips or appliques are provided by FIGS. 10A, 10B and 10C. The parameters of embodiments of the invention including the ratios of the cuticle height to apex height, apex length to total length as well as apex height, cuticle height, the apex thickness, the cuticle thickness, distal end thickness and the no-rock arrangement of the edges deliver natural appearing plastic extension embodiments that are attractive and non-artificial in appearance, fit completely on the natural nail and do not suffer from incidental pop-off. The comparisons to the closest commercial tips show that the foregoing parameters of embodiments of the plastic extension according to the invention differ from those of the commercial tips.

[0202] The differences are apparent in the flex test results shown in Table 4 and graphically represented by FIGS. 10A, 10B and 10C, Graphs A, B and C. The cuticle end flex pressures for flattening the cuticle ends of embodiments of the plastic extension according to the invention are significantly lower than those for the commercial tips. The examples show that the embodiments of the plastic extension (designated on the Figures as OPI) has a flex resistance of less than 350 g/mm, preferably less than 300 g/mm. By comparison, the commercial tips have a flex resistance about 2 to 3 times greater than the flex resistance of the embodiments of the plastic extension when similar width samples are compared. These lower flex pressures enable the embodiments of the plastic extension of the invention to spread fully over the surface of the nail plate. The lack of void space between the nail plate and the embodiment renders the embodiment of the plastic extension significantly less susceptible to accidental pop-off. The location of the apex of the embodiment of the plastic extension enables maintenance of this high point proximate to the center of the nail as the nail grows. In contrast, positioning the apex at the center of its length or too close to its distal end results in a mis-shaped nail extension appearance as the natural nail grows.

TABLE 1

Plastic extension Coffin overall shape embodiment according to the Invention								
Length	Width	Apex Height	Tip Height	Cuticle Height	Apex Location from Cuticle end	Tip H/ Apex H	Cuticle H/ Apex H	Apex L/ Total L
26.14	14.5	5.57	2.40	3.43	9.41	0.43	0.615	0.36
26.14	14.16	5.46	2.35	3.36	9.41	0.43	0.615	0.36
26.12	13.32	5.39	2.32	3.32	9.40	0.43	0.615	0.36
25.59	12.53	5.39	2.32	3.32	9.21	0.43	0.615	0.36
25.62	11.69	5.59	2.41	3.44	9.22	0.43	0.615	0.36
25.34	10.84	5.43	2.34	3.34	9.12	0.43	0.615	0.36
25	10.51	5.26	2.27	3.24	9.00	0.43	0.615	0.36
24.62	10.17	5.1	2.20	3.14	8.86	0.43	0.615	0.36
24	9.79	4.87	2.10	3.00	8.64	0.43	0.615	0.36
23.64	9.48	4.66	2.01	2.87	8.51	0.43	0.615	0.36
22.88	8.93	4.42	1.91	2.72	8.23	0.43	0.615	0.36
22	8.47	4.21	1.82	2.59	7.92	0.43	0.615	0.36
21.73	7.99	4.02	1.73	2.47	7.82	0.43	0.615	0.36
21.31	7.52	3.89	1.68	2.39	7.67	0.43	0.615	0.36
21.15	7.05	3.67	1.58	2.26	7.61	0.43	0.615	0.36
20.59	6.06	3.47	1.50	2.14	7.41	0.43	0.615	0.36

TABLE 2

Sculpted Commercial Tip								
Length	Width	Apex Height	Tip Height	Cuticle Height	Apex Location from Cuticle end	Tip Height/ Total Height	Cuticle Height/ Total Height	Apex Length/ Total Length
26.1	13.75	5.45	3.1	2.92	12.48	0.57	0.54	0.48
23.7	10.94	5.45	3.1	2.91	11.46	0.57	0.53	0.48
22.4	9.05	5.15	2.85	2.76	10.76	0.55	0.54	0.48
20.94	6.6	3.8	2.04	2.03	10.03	0.54	0.53	0.48

TABLE 3

Natural Commercial Tip								
Length	Width	Apex Height	Tip Height	Cuticle Height	Apex Location from Cuticle end	Tip Height/ Total Height	Cuticle Height/ Total Height	Apex Length/ Total Length
26.5	13.4	5.45	2.24	4	10.09	0.41	0.73	0.38
24.6	11.2	5.05	2.12	3.7	9.57	0.42	0.73	0.39
22.6	9.02	4.2	1.76	3.07	8.6	0.42	0.73	0.38
21	7.06	3.85	1.6	2.83	8.32	0.42	0.74	0.40

Dependent Parameters	Inv Embod.	Comm. Sculpted	Comm. Natural	Benefit/Relevance
Tip Height	0.43	0.56	0.42	Signifies the slope of the nail tip - Natural and Premium Look
Apex Height				
Cuticle Height	0.61	0.53	0.73	Signifies the location of the cuticle - affects fit, comfort and look
Apex Height				
Apex Length	0.36	0.48	0.39	Position of Apex affects look and structural integrity. Having apex
Total Length				

-continued				
Dependent Parameters	Inv Embod.	Comm. Sculpted	Comm. Natural	Benefit/Relevance
				too far ahead makes tips front heavy on grow out and can lead to lifting

Flex Test Table 4
Flex Tests of Cuticle Ends of Plastic Extension Embodiments of the Invention and Comm. Sculpted and Comm. Natural

[0203] Texture Analyzer Device Parameters: Test Speed: 1 mm/sec; Depth 1.5 mm;

[0204] Point of testing: Cuticle end

[0205] Stiffness measured between Displacement of 0 and 0.5 mm

Cuticle End Test

Plastic Extension Embodiment of Invention		Comm. Sculpted Tips		Comm. Natural Tips	
Size	Stiffness (g/mm)	Size	Stiffness (g/mm)	Size	Stiffness (g/mm)
D0	195	00	416	00	430
6	317	5	726	5	525
9	284	8	608	8	460

Graphs of Flex Test of OPI Cuticles and Comm. Cuticles (Cuticle Ends), FIGS. 10A, 10B, 10C

[0206] The graphs, FIGS. 10A, 10B and 10C present the flex tests of the cuticle ends of three sizes 0/00, 6/5 and 13/8 (widths) of plastic extension embodiments (OPI) and the Comm. Sculpted and Comm. Natural tips (appliques for fingernails). The bottom two plots (Graph A) three plots (Graph B) and four plots (Graph C) show respectively the Stiffness of the cuticle ends of the plastic extension embodiments of sizes 0, 6 and 13. The middle three plots of Graphs A and B and middle four plots of Graph C show the Stiffness of the cuticle ends of Comm. Natural tips. The top three plots of Graphs A, B and C show the Stiffness of the cuticle ends of Comm. Sculpted tips.

[0207] The plots of the plastic extension embodiments (OPI) show a uniform increase in slope indicating elastic deformation until plastic deformation indicated by the sharp descending curve. The conversion of the plastic extension embodiments to plastic deformation is the point at which the plastic extension embodiment cuticle ends can no longer recover their original shapes through elastic deformation. The plastic deformation point is a desirable result as it corresponds to the point at which the cuticle end will remain flattened against the nail plate without continued application of pressure. The low force required to achieve this result for the plastic extension embodiments means that a user will be able to obtain easily uniform, full surface to surface bonding between the surfaces of the plastic extension embodiments and the nail plates.

[0208] The plots of the commercial (Comm.) appliques demonstrate that a significantly higher force is required to cause plastic deformation relative to the plastic extension embodiments. Because of this higher force requirement required to flatten the commercial appliques against the nail plates, a user will have difficulty in obtaining uniform, full surface to surface bonds of the Comm. appliques and the nail plates. Some of the Comm. appliques also demonstrate strain hardening during their plastic deformation. The strain hardening is indicated by a sharp increase in the ascending slope of the plot of Stiffness before plastic deformation occurs. Strain hardening increases the difficulty of the user to be able to press the cuticle end of the Comm. appliques and flatten them against the nail plates.

Complete Plastic Extension Sets

[0209] Parameters for a complete set of plastic extension embodiments of the invention for several primary overall shapes spanning the short, medium and long length categories and sizes (widths) from D0 to 9 (size 0 to 15 with widths from 6 to 15 mm) were computed through scaling and extrapolation of the parameters, lengths and heights of the medium length category coffin shaped plastic extension embodiments described above. This complete set of parameters were used to prepare two part molds for this complete set. The two part molds were employed with acrylate copolymer pellets to form physically the complete set. The complete set included round, square, coffin, almond and stiletto overall shapes in short, medium and long length categories.

[0210] The following Detail Tables chart the parameters for this complete set of embodiments of the plastic extension including their size, length, width, apex height, (distal) tip height, cuticle height, apex location (apex length) from cuticle end, the ratio of (distal) tip height to apex height, the ratio of cuticle height to apex height, and apex length to total length. Tables H1-H5 present the short length category for five overall shapes. Tables I1-I5 present the medium length category for five overall shapes. Tables J1-J5 present the long length category for five overall shapes.

Detail Tables

[0211] Tables H1-H5 summarize parameters for the short length category for these primary round, square, coffin, almond and stiletto overall shapes. Derivatives of these short primary overall shapes may be prepared by comparison with the closest primary overall shape and longitudinal length and use of the corresponding ratios of apex length to total length, cuticle height to apex height and tip (distal) end height to apex height.

TABLE H1

Short Round						
Length	Width	Apex Height	Cuticle Height	Apex Location	Cuticle	Apex L/ Total L
				from Cuticle end	H/Apex H	
19.51	14.37	5.57	2.96	9.50	0.531	0.49
19.48	14.21	5.47	2.91	9.49	0.531	0.49
19.51	13.33	5.41	2.87	9.50	0.531	0.49
19.33	12.52	5.39	2.86	9.42	0.531	0.49
19.41	11.67	5.57	2.96	9.44	0.531	0.49
19.04	10.8	5.37	2.85	9.28	0.531	0.49
18.69	10.53	5.25	2.79	9.10	0.531	0.49
18.32	10.17	5.08	2.70	8.92	0.531	0.49
17.99	9.77	4.9	2.60	8.76	0.531	0.49
17.71	9.51	4.66	2.48	8.63	0.531	0.49
17.06	8.93	4.7	2.50	8.31	0.531	0.49
16.66	8.47	4.24	2.25	8.11	0.531	0.49
16.35	7.95	3.96	2.10	7.96	0.531	0.49
16.08	7.5	3.85	2.05	7.83	0.531	0.49
15.74	7.04	3.66	1.94	7.67	0.531	0.49
15.47	6.15	3.46	1.84	7.54	0.531	0.49

TABLE H2

Short Square						
Length	Width	Apex Height	Cuticle Height	Apex Location from Cuticle end	Cuticle H/Apex H	Apex L/Total L
20.41	14.43	5.63	3.09	9.86	0.549	0.48
20.30	14.23	5.55	3.05	9.81	0.549	0.48
20.28	13.39	5.49	3.01	9.79	0.549	0.48
20.06	12.57	5.44	2.99	9.69	0.549	0.48
20.06	11.75	5.63	3.09	9.69	0.549	0.48
19.64	10.94	5.45	2.99	9.49	0.549	0.48
19.24	10.55	5.28	2.90	9.29	0.549	0.48
18.82	10.22	5.13	2.82	9.09	0.549	0.48
18.46	9.83	4.89	2.68	8.91	0.549	0.48
18.15	9.53	4.68	2.57	8.76	0.549	0.48
17.44	8.98	4.44	2.44	8.42	0.549	0.48
16.99	8.5	4.22	2.32	8.20	0.549	0.48
16.66	8.03	4.03	2.21	8.05	0.549	0.48
16.32	7.55	3.89	2.14	7.88	0.549	0.48
15.97	7.08	3.67	2.01	7.71	0.549	0.48
15.67	6.1	3.47	1.90	7.57	0.549	0.48

TABLE H3

Short Coffin						
Length	Width	Apex Height	Cuticle Height	Apex Location from Cuticle end	Cuticle H/Apex H	Apex L/Total L
20.57	14.5	5.57	3.43	9.41	0.615	0.46
20.57	14.16	5.46	3.36	9.41	0.615	0.46
20.55	13.32	5.39	3.32	9.40	0.615	0.46
20.13	12.53	5.39	3.32	9.21	0.615	0.46
20.16	11.69	5.59	3.44	9.22	0.615	0.46
19.94	10.84	5.43	3.34	9.12	0.615	0.46
19.67	10.51	5.26	3.24	9.00	0.615	0.46
19.37	10.17	5.1	3.14	8.86	0.615	0.46
18.88	9.79	4.87	3.00	8.64	0.615	0.46
18.60	9.48	4.66	2.87	8.51	0.615	0.46
18.00	8.93	4.42	2.72	8.23	0.615	0.46
17.31	8.47	4.21	2.59	7.92	0.615	0.46
17.10	7.99	4.02	2.47	7.82	0.615	0.46
16.77	7.52	3.89	2.39	7.67	0.615	0.46
16.64	7.05	3.67	2.26	7.61	0.615	0.46
16.20	6.06	3.47	2.14	7.41	0.615	0.46

TABLE H4

Short Almond						
Length	Width	Apex Height	Cuticle Height	Apex Location from Cuticle end	Cuticle H/Apex H	Apex L/Total L
20.27	14.39	5.6	3.44	9.01	0.614	0.44
20.21	14.18	5.47	3.36	8.98	0.614	0.44

TABLE H4-continued

Short Almond						
Length	Width	Apex Height	Cuticle Height	Apex Location from Cuticle end	Cuticle H/Apex H	Apex L/Total L
20.27	13.31	5.39	3.31	9.01	0.614	0.44
20.12	12.52	5.39	3.31	8.94	0.614	0.44
20.18	11.65	5.6	3.44	8.97	0.614	0.44
19.80	10.83	5.4	3.32	8.8	0.615	0.44
19.45	10.46	5.27	3.24	8.65	0.615	0.44
19.09	10.14	5.09	3.13	8.48	0.615	0.44
18.77	9.79	4.89	3	8.34	0.613	0.44
18.49	9.46	4.67	2.87	8.22	0.615	0.44
17.83	8.94	4.41	2.71	7.92	0.615	0.44
17.42	8.47	4.22	2.59	7.74	0.614	0.44
17.13	7.99	4.02	2.47	7.61	0.614	0.44
16.83	7.52	3.89	2.39	7.48	0.614	0.44
16.52	7.05	3.7	2.27	7.34	0.614	0.44
16.25	6.16	3.5	2.15	7.22	0.614	0.44

TABLE H5

Short Stiletto						
Length	Width	Apex Height	Cuticle Height	Apex Location from Cuticle end	Cuticle H/Apex Total H	Apex L/Total L
20.84	14.46	5.56	3.42	8.92	0.615	0.43
20.81	14.14	5.43	3.34	8.91	0.615	0.43
20.86	13.32	5.41	3.33	8.93	0.615	0.43
20.72	12.52	5.38	3.31	8.86	0.615	0.43
20.78	11.66	5.58	3.43	8.89	0.615	0.43
20.43	10.88	5.39	3.31	8.74	0.615	0.43
20.10	10.55	5.24	3.22	8.60	0.615	0.43
19.76	10.17	5.13	3.15	8.46	0.615	0.43
19.46	9.75	4.85	2.98	8.33	0.615	0.43
19.22	9.5	4.67	2.87	8.22	0.615	0.43
18.61	8.92	4.43	2.72	7.96	0.615	0.43
18.23	8.47	4.24	2.61	7.80	0.615	0.43
17.95	8.01	4.04	2.48	7.68	0.615	0.43
17.70	7.57	3.9	2.40	7.57	0.615	0.43
17.40	7.05	3.68	2.26	7.45	0.615	0.43
17.15	6.08	3.48	2.14	7.34	0.615	0.43

Tables I1-I5 summarize parameters for the medium length category for these primary round, square, coffin, almond and stiletto overall shapes. Derivatives of these medium primary overall shapes may be prepared by comparison with the closest primary overall shape and longitudinal length and use of the corresponding ratios of apex length to total length, cuticle height to apex height and tip (distal) end height to apex-height.

TABLE I1

Round								
Length	Width	Apex Height	Tip Height	Cuticle Height	Apex Location from Cuticle end	Tip H/ Apex H	Cuticle H/ Apex H	Apex L/ Total L
24.26	14.37	5.57	2.69	2.96	9.50	0.48	0.531	0.39
24.22	14.21	5.47	2.64	2.91	9.49	0.48	0.531	0.39
24.26	13.33	5.41	2.61	2.87	9.50	0.48	0.531	0.39
24.04	12.52	5.39	2.60	2.86	9.42	0.48	0.531	0.39
24.14	11.67	5.57	2.69	2.96	9.44	0.48	0.531	0.39
23.68	10.8	5.37	2.59	2.85	9.28	0.48	0.531	0.39
23.24	10.53	5.25	2.54	2.79	9.10	0.48	0.531	0.39
22.78	10.17	5.08	2.45	2.70	8.92	0.48	0.531	0.39
22.37	9.77	4.9	2.37	2.60	8.76	0.48	0.531	0.39
22.02	9.51	4.66	2.25	2.48	8.63	0.48	0.531	0.39
21.21	8.93	4.7	2.27	2.50	8.31	0.48	0.531	0.39
20.71	8.47	4.24	2.05	2.25	8.11	0.48	0.531	0.39
20.33	7.95	3.96	1.91	2.10	7.96	0.48	0.531	0.39
19.99	7.5	3.85	1.86	2.05	7.83	0.48	0.531	0.39
19.57	7.04	3.66	1.77	1.94	7.67	0.48	0.531	0.39
19.24	6.15	3.46	1.67	1.84	7.54	0.48	0.531	0.39

TABLE I2

Square								
Length	Width	Apex Height	Tip Height	Cuticle Height	Apex Location from Cuticle end	Tip H/ Apex H	Cuticle H/ Apex H	Apex L/ Total L
25.55	14.43	5.63	3.16	3.09	9.86	0.56	0.549	0.39
25.42	14.23	5.55	3.12	3.05	9.81	0.56	0.549	0.39
25.39	13.39	5.49	3.08	3.01	9.79	0.56	0.549	0.39
25.12	12.57	5.44	3.05	2.99	9.69	0.56	0.549	0.39
25.12	11.75	5.63	3.16	3.09	9.69	0.56	0.549	0.39
24.59	10.94	5.45	3.06	2.99	9.49	0.56	0.549	0.39
24.09	10.55	5.28	2.96	2.90	9.29	0.56	0.549	0.39
23.57	10.22	5.13	2.88	2.82	9.09	0.56	0.549	0.39
23.11	9.83	4.89	2.74	2.68	8.91	0.56	0.549	0.39
22.72	9.53	4.68	2.63	2.57	8.76	0.56	0.549	0.39
21.84	8.98	4.44	2.49	2.44	8.42	0.56	0.549	0.39
21.27	8.5	4.22	2.37	2.32	8.20	0.56	0.549	0.39
20.86	8.03	4.03	2.26	2.21	8.05	0.56	0.549	0.39
20.44	7.55	3.89	2.18	2.14	7.88	0.56	0.549	0.39
20	7.08	3.67	2.06	2.01	7.71	0.56	0.549	0.39
19.62	6.1	3.47	1.95	1.90	7.57	0.56	0.549	0.39

TABLE I3

Coffin								
Length	Width	Apex Height	Tip Height	Cuticle Height	Apex Location from Cuticle end	Tip H/ Apex H	Cuticle H/ Apex H	Apex L/ Total L
26.14	14.5	5.57	2.40	3.43	9.41	0.43	0.615	0.36
26.14	14.16	5.46	2.35	3.36	9.41	0.43	0.615	0.36
26.12	13.32	5.39	2.32	3.32	9.40	0.43	0.615	0.36
25.59	12.53	5.39	2.32	3.32	9.21	0.43	0.615	0.36
25.62	11.69	5.59	2.41	3.44	9.22	0.43	0.615	0.36
25.34	10.84	5.43	2.34	3.34	9.12	0.43	0.615	0.36
25	10.51	5.26	2.27	3.24	9.00	0.43	0.615	0.36
24.62	10.17	5.1	2.20	3.14	8.86	0.43	0.615	0.36
24	9.79	4.87	2.10	3.00	8.64	0.43	0.615	0.36
23.64	9.48	4.66	2.01	2.87	8.51	0.43	0.615	0.36

TABLE I3-continued

Coffin								
Length	Width	Apex Height	Tip Height	Cuticle Height	Apex Location from Cuticle end	Tip H/ Apex H	Cuticle H/ Apex H	Apex L/ Total L
22.88	8.93	4.42	1.91	2.72	8.23	0.43	0.615	0.36
22	8.47	4.21	1.82	2.59	7.92	0.43	0.615	0.36
21.73	7.99	4.02	1.73	2.47	7.82	0.43	0.615	0.36
21.31	7.52	3.89	1.68	2.39	7.67	0.43	0.615	0.36
21.15	7.05	3.67	1.58	2.26	7.61	0.43	0.615	0.36
20.59	6.06	3.47	1.50	2.14	7.41	0.43	0.615	0.36

TABLE I4

Almond								
Length	Width	Apex Height	Tip Height	Cuticle Height	Apex Location from Cuticle end	Tip H/ Apex H	Cuticle H/ Apex H	Apex L/ Total L
25.23	14.39	5.6	2.43	3.44	9.01	0.43	0.614	0.36
25.16	14.18	5.47	2.37	3.36	8.98	0.43	0.614	0.36
25.23	13.31	5.39	2.34	3.31	9.01	0.43	0.614	0.36
25.04	12.52	5.39	2.34	3.31	8.94	0.43	0.614	0.36
25.12	11.65	5.6	2.43	3.44	8.97	0.43	0.614	0.36
24.65	10.83	5.4	2.34	3.32	8.8	0.43	0.615	0.36
24.21	10.46	5.27	2.29	3.24	8.65	0.43	0.615	0.36
23.76	10.14	5.09	2.21	3.13	8.48	0.43	0.615	0.36
23.36	9.79	4.89	2.12	3	8.34	0.43	0.613	0.36
23.01	9.46	4.67	2.03	2.87	8.22	0.43	0.615	0.36
22.19	8.94	4.41	1.91	2.71	7.92	0.43	0.615	0.36
21.68	8.47	4.22	1.83	2.59	7.74	0.43	0.614	0.36
21.32	7.99	4.02	1.74	2.47	7.61	0.43	0.614	0.36
20.95	7.52	3.89	1.69	2.39	7.48	0.43	0.614	0.36
20.56	7.05	3.7	1.61	2.27	7.34	0.44	0.614	0.36
20.23	6.16	3.5	1.52	2.15	7.22	0.43	0.614	0.36

TABLE I5

Stiletto								
Length	Width	Apex Height	Tip Height	Cuticle Height	Apex Location from Cuticle end	Tip H/ Apex H	Cuticle H/ Apex H	Apex L/ Total L
28.09	14.46	5.56	1.76	3.42	8.92	0.32	0.615	0.32
28.05	14.14	5.43	1.72	3.34	8.91	0.32	0.615	0.32
28.12	13.32	5.41	1.72	3.33	8.93	0.32	0.615	0.32
27.92	12.52	5.38	1.71	3.31	8.86	0.32	0.615	0.32
28	11.66	5.58	1.77	3.43	8.89	0.32	0.615	0.32
27.53	10.88	5.39	1.71	3.31	8.74	0.32	0.615	0.32
27.09	10.55	5.24	1.66	3.22	8.60	0.32	0.615	0.32
26.63	10.17	5.13	1.63	3.15	8.46	0.32	0.615	0.32
26.23	9.75	4.85	1.54	2.98	8.33	0.32	0.615	0.32
25.9	9.5	4.67	1.48	2.87	8.22	0.32	0.615	0.32
25.08	8.92	4.43	1.41	2.72	7.96	0.32	0.615	0.32
24.57	8.47	4.24	1.34	2.61	7.80	0.32	0.615	0.32
24.19	8.01	4.04	1.28	2.48	7.68	0.32	0.615	0.32
23.85	7.57	3.9	1.24	2.40	7.57	0.32	0.615	0.32
23.45	7.05	3.68	1.17	2.26	7.45	0.32	0.615	0.32
23.12	6.08	3.48	1.10	2.14	7.34	0.32	0.615	0.32

Tables J1-J5 summarizing parameters for the long length category for these primary round, square, coffin, almond and stiletto overall shapes. Derivatives of these long primary overall shapes may be prepared by comparison with the closest primary overall shape and longitudinal length and use of the corresponding ratios of apex length to total length, cuticle height to apex height and tip (distal) end height to apex height.

TABLE J1

Round						
Length	Width	Apex Height	Cuticle Height	Apex Location from Cuticle end	Cuticle H/Apex H	Apex L/ Total L
30.65	14.37	5.57	2.96	9.50	0.531	0.31
30.60	14.21	5.47	2.91	9.49	0.531	0.31
30.65	13.33	5.41	2.87	9.50	0.531	0.31
30.38	12.52	5.39	2.86	9.42	0.531	0.31
30.45	11.67	5.57	2.96	9.44	0.531	0.31
29.92	10.8	5.37	2.85	9.28	0.531	0.31
29.36	10.53	5.25	2.79	9.10	0.531	0.31
28.78	10.17	5.08	2.70	8.92	0.531	0.31
28.27	9.77	4.9	2.60	8.76	0.531	0.31
27.82	9.51	4.66	2.48	8.63	0.531	0.31
26.80	8.93	4.7	2.50	8.31	0.531	0.31
26.17	8.47	4.24	2.25	8.11	0.531	0.31
25.69	7.95	3.96	2.10	7.96	0.531	0.31
25.26	7.5	3.85	2.05	7.83	0.531	0.31
24.73	7.04	3.66	1.94	7.67	0.531	0.31
24.31	6.15	3.46	1.84	7.54	0.531	0.31

TABLE J2

Square						
Length	Width	Apex Height	Cuticle Height	Apex Location from Cuticle end	Cuticle H/Apex H	Apex L/ Total L
31.79	14.43	5.63	3.09	9.86	0.549	0.31
31.63	14.23	5.55	3.05	9.81	0.549	0.31
31.59	13.39	5.49	3.01	9.79	0.549	0.31
31.26	12.57	5.44	2.99	9.69	0.549	0.31
31.26	11.75	5.63	3.09	9.69	0.549	0.31
30.60	10.94	5.45	2.99	9.49	0.549	0.31
29.98	10.55	5.28	2.90	9.29	0.549	0.31
29.33	10.22	5.13	2.82	9.09	0.549	0.31
28.76	9.83	4.89	2.68	8.91	0.549	0.31
28.27	9.53	4.68	2.57	8.76	0.549	0.31
27.18	8.98	4.44	2.44	8.42	0.549	0.31
26.47	8.5	4.22	2.32	8.20	0.549	0.31
25.96	8.03	4.03	2.21	8.05	0.549	0.31
25.43	7.55	3.89	2.14	7.88	0.549	0.31
24.89	7.08	3.67	2.01	7.71	0.549	0.31
24.41	6.1	3.47	1.90	7.57	0.549	0.31

TABLE J3

Coffin						
Length	Width	Apex Height	Cuticle Height	Apex Location from Cuticle end	Cuticle H/Apex H	Apex L/ Total L
31.36	14.5	5.57	3.43	9.41	0.615	0.30
31.36	14.16	5.46	3.36	9.41	0.615	0.30
31.33	13.32	5.39	3.32	9.40	0.615	0.30
30.70	12.53	5.39	3.32	9.21	0.615	0.30
30.73	11.69	5.59	3.44	9.22	0.615	0.30

TABLE J3-continued

Coffin						
Length	Width	Apex Height	Cuticle Height	Apex Location from Cuticle end	Cuticle H/Apex H	Apex L/ Total L
30.40	10.84	5.43	3.34	9.12	0.615	0.30
29.99	10.51	5.26	3.24	9.00	0.615	0.30
29.53	10.17	5.1	3.14	8.86	0.615	0.30
28.79	9.79	4.87	3.00	8.64	0.615	0.30
28.36	9.48	4.66	2.87	8.51	0.615	0.30
27.45	8.93	4.42	2.72	8.23	0.615	0.30
26.39	8.47	4.21	2.59	7.92	0.615	0.30
26.07	7.99	4.02	2.47	7.82	0.615	0.30
25.56	7.52	3.89	2.39	7.67	0.615	0.30
25.37	7.05	3.67	2.26	7.61	0.615	0.30
24.70	6.06	3.47	2.14	7.41	0.615	0.30

TABLE J4

Almond						
Length	Width	Apex Height	Cuticle Height	Apex Location from Cuticle end	Cuticle H/Apex H	Apex L/ Total L
32.18	14.39	5.6	3.44	9.01	0.614	0.28
32.07	14.18	5.47	3.36	8.98	0.614	0.28
32.18	13.31	5.39	3.31	9.01	0.614	0.28
31.93	12.52	5.39	3.31	8.94	0.614	0.28
32.04	11.65	5.6	3.44	8.97	0.614	0.28
31.43	10.83	5.4	3.32	8.8	0.615	0.28
30.89	10.46	5.27	3.24	8.65	0.615	0.28
30.29	10.14	5.09	3.13	8.48	0.615	0.28
29.79	9.79	4.89	3	8.34	0.613	0.28
29.36	9.46	4.67	2.87	8.22	0.615	0.28
28.29	8.94	4.41	2.71	7.92	0.615	0.28
27.64	8.47	4.22	2.59	7.74	0.614	0.28
27.18	7.99	4.02	2.47	7.61	0.614	0.28
26.71	7.52	3.89	2.39	7.48	0.614	0.28
26.21	7.05	3.7	2.27	7.34	0.614	0.28
25.79	6.16	3.5	2.15	7.22	0.614	0.28

TABLE J1

Stiletto						
Length	Width	Apex Height	Cuticle Height	Apex Location from Cuticle end	Cuticle H/Apex H	Apex L/ Total L
33.03	14.46	5.56	3.42	8.92	0.615	0.27
32.98	14.14	5.43	3.34	8.91	0.615	0.27
33.07	13.32	5.41	3.33	8.93	0.615	0.27
32.83	12.52	5.38	3.31	8.86	0.615	0.27
32.93	11.66	5.58	3.43	8.89	0.615	0.27
32.37	10.88	5.39	3.31	8.74	0.615	0.27
31.86	10.55	5.24	3.22	8.60	0.615	0.27
31.31	10.17	5.13	3.15	8.46	0.615	0.27
30.84	9.75	4.85	2.98	8.33	0.615	0.27
30.46	9.5	4.67	2.87	8.22	0.615	0.27
29.49	8.92	4.43	2.72	7.96	0.615	0.27
28.89	8.47	4.24	2.61	7.80	0.615	0.27
28.45	8.01	4.04	2.48	7.68	0.615	0.27
28.05	7.57	3.9	2.40	7.57	0.615	0.27
27.58	7.05	3.68	2.26	7.45	0.615	0.27
27.19	6.08	3.48	2.14	7.34	0.615	0.27

Working Example

Fitting a Plastic Extension Embodiment and a Commercial Tip to a Test Subject

[0212] A size 1 overall coffin shape 19 mm length example of the plastic extension embodiment and a competitor commercial tip were sized to the test subject's right ring finger and right middle finger respectively. Each nail was coated with a tacky surface of gel on each nail. The coffin shaped plastic extension embodiment and the competitor commercial coffin tip were placed in the respective fingers of the test subject, positioned and pressed onto the sticky gel. The fitted plastic extension embodiment and fitted competitor tip were irradiated with full spectrum UV to cure the gel fully and adhere the plastic extension and competitor tip to the ring fingernail and middle fingernail respectively. The ends of both of the plastic extension embodiment and the competitor tip were trimmed with a curved nail clipper to provide rounded ends.

[0213] Upon flexing each finger with the attached plastic extension embodiment and competitor tip, the test subject remarked that the attached plastic extension embodiment felt comfortable. He said that he did not feel a pull upward and forward on his fingernail. In contrast, the test subject said that the competitor tip pulled up on his nail and made the nail feel tight. While the test subject had never worn artificial nails or any other nail covering before this test, he said that the plastic extension embodiment appeared to look like the natural nail of his other hand while the competitor tip appeared distinctively different and artificial compared with the natural nail of his other hand.

[0214] Photographs of the attached competitor tip and the example of the plastic extension embodiment are shown as FIGS. 11A and 11B respectively. The competitor tip photo 11A shows an apex further ahead of the natural position so that the overall structure appears to be rising. Reducing the length by clipping the end to provide a rounded end left the look of the attached competitor tip incomplete. The plastic extension embodiment photo 11B shows an optimized location of the apex relative to the natural apex of the nail. Clipping the end to provide a rounded end does not affect the apex or natural look of the attached plastic extension embodiment.

[0215] A fingertip photograph of the two fingers with attached plastic extension embodiment and commercial tip is shown as FIG. 12. The competitor tip end-on view on the middle finger shows a rigid fit, a flattened apex on the nail bed and a big gap between the natural nail and the tip. The plastic extension embodiment end-on view on the ring finger shows a raised apex on the nail bed and a snug/comfort fit with less gap between the natural nail and the plastic extension embodiment.

THE STATEMENTS OF EMBODIMENTS OF THE INVENTION

[0216] The Statements of the Invention set forth further Embodiments of the Invention and provide details described above as well as additional details according to the Invention. The Statements supplement and expand the description of the plastic extension embodiments set forth in the

Detailed Description. Discrepancies between the Statements and Detailed Description are to be considered to be additive rather than subtractive.

Statement 1A.

1A. A plastic extension for covering a nail plate comprising:

- [0217] a) a top surface, a bottom surface, a cuticle end, a distal end, a left edge and a right edge;
- [0218] b) a longitudinal length of from about 15 mm to about 37 mm as a straight line from the midpoint of cuticle end to the midpoint of the distal end and running along a presumed horizontal plane between the left edge and right edge and the longitudinal length is divided into categories of short, medium and long;
- [0219] c) a width between left and right edges of from about 6 mm to about 15 mm wherein the width is measured at the widest distance between the left and right edges;
- [0220] d) a longitudinal curve running from the cuticle end to the distal end which increases in slope from the cuticle end to an apex position (602) and then decreases in slope from the apex to the distal end, and a latitudinal curve extending between the left and right edges and running from the cuticle end to the distal end;
- [0221] e) an apex point (606) on the longitudinal length which is directly coincident with and below the apex position (602) of the longitudinal curve wherein the length between the cuticle end of the longitudinal length and the apex point (606) is the apex length (607);
- [0222] f) a ratio of the apex length to the longitudinal length of from 0.27 to 0.49 which ratio determines the apex length according to an inverse relationship of the ratio and the longitudinal length;
- [0223] g) an apex height of from about 3 mm to about 6 mm for all length categories as measured vertically from the top surface at the apex to a presumed horizontal plane between the left and right edges positioned directly below the apex;
- [0224] h) a cuticle height of from about 1.5 mm to about 3.5 mm relative to the presumed horizontal plane between the left and right edges;
- [0225] i) a ratio of the cuticle height to apex height of about 0.5 to about 0.62;
- [0226] j) a distal end thickness of no more than about 1 mm;
- [0227] k) a cuticle end thickness of no more than about 1 mm;
- [0228] l) an apex thickness of no more than about 1.5 mm.

Statement 1B

1B. A plastic extension having an overall shape for covering a nail plate comprising:

- [0229] a) a top surface, a bottom surface, a cuticle end, a distal end, a left edge and a right edge;
- [0230] b) a longitudinal length of from about 15.0 mm to about 35.0 mm as a straight line from the midpoint of cuticle end to the midpoint of the distal end and running along a presumed horizontal plane between the left edge and right edge;
- [0231] c) longitudinal length categories of short ranging from about 15 mm to about 21 mm, medium ranging

from about 19 mm to about 28 mm and long ranging from about 24 mm to about 35 mm;

[0232] d) a longitudinal curve running from the cuticle end to the distal end which increases in slope from the cuticle end to an apex position (602) and then decreases in slope from the apex to the distal end, and a latitudinal curve extending between the left and right edges from the cuticle end to the distal end;

[0233] e) a width between left and right edges of from about 6 mm to about 15 mm wherein the width is measured at the widest distance between the left and right edges;

[0234] f) an apex point (606) on the longitudinal length which is directly coincident with and below the apex position (602) of the longitudinal curve wherein the length between the cuticle end of the longitudinal length and the apex point (606) is the apex length (607);

[0235] g) a ratio of the apex length to the longitudinal length coordinated by the length category and the overall shape as provided by the following Table B

TABLE B

Length Category	Length in mm	Apex to Length Ratio
Short	15-21	0.41-0.49
Medium	19-28	0.32-0.40
Long	24-37	0.27-0.31

[0236] h) an apex height of from about 3 mm to about 6 mm measured vertically from the top surface at the apex to a presumed horizontal plane between the left and right edges positioned directly below the apex;

[0237] i) a cuticle height of from about 1.5 mm to about 3.5 mm relative to the presumed horizontal plane between the left and right edges;

[0238] j) a ratio of the cuticle height to apex height of about 0.5 to about 0.62;

[0239] k) a distal end thickness of no more than about 1 mm;

[0240] l) a cuticle end thickness of no more than about 1 mm;

[0241] m) an apex thickness of no more than about 1.5 mm.

[0242] 1C. The plastic extension of statement 1A or 1B further comprising a distal end height of no more than about 4.5 mm, preferably about 0.5 mm to about 4.5 mm, more preferably about 0.5 mm to about 4 mm, more preferably about 0.5 mm to about 3.5 mm, especially more preferably about 1 mm to about 3.5 mm as measured at the point along the length of the plastic extension indicated by the comparison with the distal end of a similar plastic extension of the medium length category.

[0243] 1D. The plastic extension of statement 1A, 1B or 1C wherein the distal end thickness is from about 0.6 mm to about 0.9 mm; the cuticle end thickness is from about 0.2 mm to about 0.4 mm; and the apex thickness is from about 0.6 mm to about 0.7 mm.

[0244] 1E. The plastic extension of statement 1A, 1B, 1C or 1C wherein the length categories are segmented by width ranges according to Table A:

TABLE A

Avg. Width (mm)	Short (mm)	Medium (mm)	Long (mm)
12-15	19-21	22-28	29-37
9.5-12	18-21	22-28	29-34
8-9.5	16-20	21-26	27-32
6-8	15-18	19-23	24-30

[0245] 2. The plastic extension of statement 1A, 1B, 1C, 1D or 1E which is stable and does not move when placed on a flat surface with the left and right edges of the plastic extension in contact with the flat surface and a downward force is applied to the cuticle end or the distal end of the plastic extension.

[0246] 3. The plastic extension according to any of the preceding statements further comprising members of a group of plastic extensions having one overall shape wherein each member has a width of one of 15 increments selected from the width range of from about 6 mm to about 15 mm and a length selected according to the overall shape.

[0247] 4. The members of the group of plastic extensions according to statement 3 wherein the widths of the members are selected according to the widths of the thumb and fingers of a hand onto which the plastic extensions are to be placed.

[0248] 5. The plastic extension according to any of the preceding statements wherein the bottom surface of the cuticle end meshes with the cuticle surface of a fingernail when the plastic extension is placed on a fingernail and downward finger pressure is applied to the cuticle end.

[0249] 6. The plastic extension according to statement 5 wherein the finger pressure applied to the cuticle end of the plastic extension provides essentially complete contact between the natural nail and the bottom surface of the cuticle end.

[0250] 7. The plastic extension according to statement 4 comprising a collection of groups for all sizes of hands.

[0251] 8. The plastic extension according to any of the preceding statements wherein the apex height is about 3.2 mm to about 5.7 mm.

[0252] 9. The plastic extension according to any of the preceding statements wherein the ratio of cuticle height to apex height is about 0.53 to about 0.62.

[0253] 10. The plastic extension according to any of the preceding statements wherein the distal end thickness is about 0.68 mm to about 0.89 mm.

[0254] 12. The plastic extension according to any of the preceding statements wherein the cuticle end thickness is about 0.2 mm to about 0.35 mm.

[0255] 13. The plastic extension according to any of the preceding statements wherein the apex thickness is about 0.61 mm to about 0.68 mm.

[0256] 14. A nail applique kit comprising an adhesive and one or more plastic extensions according to any of the preceding statements.

[0257] 15. The nail applique kit according to statement 14 wherein the adhesive is a UV curable gel, a cyanoacrylate adhesive, a free radical curable composition, a pressure sensitive adhesive or a dual coated slip.

[0258] 16. The nail applique kit according to statement 14 or 15 wherein the adhesive is a UV curable gel.

[0259] 17. The nail applique kit according to statement 16 wherein the nail applique kit further comprises a UV lamp preferably constructed of LED diodes for generation of electromagnetic radiation in the range of 280 to 420 nm.

[0260] 18. A nail applique kit according to statement 16 or 17 wherein the UV gel includes one or more UV photoinitiators which have selective excitation segments in a range of 20 nm to 60 nm within a UV radiation wavelength range of from 280 to 420 nm.

[0261] 19. The nail applique kit according to any of statements 16-18 wherein the UV lamp includes a control circuit, preferably an integrated circuit with software management, and the control circuit is capable of selective control of the wavelength of UV light generated and the time period of UV light generation.

[0262] 20. The nail applicator kit according to any of statements 16-19 wherein the selective control of the generated UV light wavelength enables flash cure of the UV gel or UVB partial cure of the UV gel to produce a tacky state of the UV gel.

[0263] 21. The nail applicator kit according to any of statement 16-20 wherein the tacky state exists for at least 5 minutes, preferably at least 10 minutes and the tacky UV gel may be fully cured with 280 to 420 nm irradiation which spans UVA and UVB irradiation.

[0264] 22. The nail applique kit according to any of statements 16-21 wherein the UV curable gel before UV cure or preferably in the flash cure or UVB cure tacky state exhibits adhesion when in contact with a nail plate and/or the bottom surface of the plastic extension.

[0265] 23. The nail applique kit according to any of statements 16-22 wherein the UV curable gel produces a temperature spike of no more than 58° C. when UV irradiated to full cure.

[0266] 24. The nail applique kit according to any of preceding statements 16-23 wherein the UV gel includes (meth)acrylate ester monomers wherein the alcohol residue of the ester is C1-C10 alkyl, C5-C10 cycloalkyl, C7-C10 aryl alkyl or a combination thereof.

[0267] 25. The nail applique kit according to any of the preceding statements 16-24 wherein the UV gel includes a (meth)acrylate dimer formed with a hydroxyalkyl (meth)acrylate as the dimer termini and a polyol, polyester, polyurethane or urethane-polyol copolymer as the linking chain between the dimer termini.

[0268] 26. The nail applique kit according to any of the preceding statements 16-25 wherein the UV gel includes an alkyl (meth)acrylate and/or a cycloalkyl (meth)acrylate and/or an arylalkyl (meth)acrylate and a (meth)acrylate dimer comprising a polymeric or oligomeric ethylene oxide terminally esterified with (meth)acrylate.

[0269] 27. The nail applique kit according to any of preceding statements 16-26 wherein the UV gel includes an alkyl (meth)acrylate and/or a cycloalkyl (meth)acrylate and/or an arylalkyl (meth)acrylate and an olefinic dimer comprising a (meth)acrylate terminated linear polyurethane.

[0270] 28. The nail applique kit according to any of preceding statements 16-27 wherein the UV gel includes an alkyl (meth)acrylate and/or a cycloalkyl (meth)acrylate and/or an arylalkyl (meth)acrylate and an olefinic dimer comprising a (meth)acrylate terminated linear polymer of a block copolymer of polyol units formed from ethylene oxide, propylene oxide or tetrahydrofuran and urethane units formed from isophorone diisocyanate or bis (isocyanato phenyl) methane and ethylene glycol, di-ethylene glycol or tri-ethylene glycol.

[0271] 29. The nail applique kit according to any of preceding statements 16-28 wherein the UV gel includes an

alkyl (meth)acrylate and/or a cycloalkyl (meth)acrylate and/or an arylalkyl (meth)acrylate and an olefinic dimer comprising a (meth)acrylate terminated linear ester polymer of a C4-C6 diacid and a C2-C6 diol.

[0272] 30. The nail applique kit according to statement 14 or 15 wherein the adhesive is a cyanoacrylate adhesive preferably including ethyl cyanoacrylate and a C4-C8 alkyl cyanoacrylate.

[0273] 31. The nail applique kit according to statement 30 wherein the ethyl cyanoacrylate is present at about 5 wt % to about 20 wt % and the C4-C8 alkyl cyanoacrylate is present at about 95 wt % to about 80 wt % relative to the total weight of the cyanoacrylate adhesive.

[0274] 32. The nail applique kit according to statement 14 or 15 wherein the adhesive is a free radical curable composition and the free radical curable monomer composition comprises one or more C1-C8 alkyl (meth)acrylate esters, an optional di or tri (meth)acrylate crosslinker, an amine accelerator, an optional inert, volatile organic solvent and an optional free radical scavenger and a separate peroxide initiator.

[0275] 33. The nail applique kit according to statement 14 or 15 wherein the adhesive is a pressure sensitive adhesive and the pressure sensitive adhesive comprises a (meth)acrylic acid/ester copolymer optionally including from about 1 wt % to about 20 wt %, preferably about 2 wt % to about 5 wt % organic solvent.

[0276] 34. The nail applique kit according to statement 14 or 15 wherein the adhesive is a dual coated slip constructed of a silicone or polyolefin sheet having the same overall shape as the plastic extension wherein with one side of the slip is coated with a pressure sensitive adhesive having a high tack and high adhesion and the other side of the slip is coated with a pressure sensitive adhesive having high tack and moderate adhesion; wherein the high tack, high adhesion side of the dual coated slip is arranged to contact the bottom surface of the plastic extension and the dual coated slip.

[0277] 35. The nail applique kit according to any of preceding statements 14-34 wherein the adhesive is contained in a unit with an accessible opening and the kit includes an applicator for applying the adhesive to a fingernail.

[0278] 36. A method using the nail applique kit of any of statements 14-35 comprising applying the adhesive to a fingernail nail plate, optionally processing the adhesive to render it at least partially adherent, fitting the plastic extension to the fingernail with adhesive, allowing the adhesive to bind the plastic extension to the fingernail nail plate.

[0279] 37. The method according to statement 36 wherein the adhesive is a UV curable gel and the UV curable gel is processed to bind the plastic extension to the fingernail plate.

[0280] 38. The method according to statement 36 wherein the adhesive is a cyanoacrylate and the cyanoacrylate is processed to bind the plastic extension to the fingernail plate.

[0281] 39. The method according to statement 36 wherein the adhesive is a free radical curable composition and the free radical curable composition is processed to bind the plastic extension to the fingernail plate.

[0282] 40. The method according to statement 36 wherein the adhesive is a pressure sensitive adhesive and the pressure sensitive adhesive is processed to bind the plastic extension to the fingernail plate.

[0283] 41. A fingernail applique comprising the plastic extension bound to a fingernail with an adhesive according to any method statements 36-40.

[0284] 42. Use of the nail applique kit according to any of statement 14-35.

[0285] 43. Use of the plastic extension of any of statements 1-13 as a fingernail applique on a fingernail.

SUMMARY STATEMENTS

[0286] The inventions, examples, results and embodiment statements described and claimed herein may have attributes and embodiments include, but not limited to, those set forth or described or referenced in this application.

[0287] All patents, publications, scientific articles, web sites and other documents and ministerial references or mentioned herein are indicative of the levels of skill of those skilled in the art to which the invention pertains, and each such referenced document and material is hereby incorporated by reference to the same extent as if it had been incorporated verbatim and set forth in its entirety herein. The right is reserved to physically incorporate into this specification any and all materials and information from any such patent, publication, scientific article, web site, electronically available information, textbook or other referenced material or document.

[0288] The written description of this patent application includes all claims and embodiment statements. All claims including all original claims as well as all embodiment statements are hereby incorporated by reference in their entirety into the written description portion of the specification and the right is reserved to physically incorporated into the written description or any other portion of the application any and all such claims. Thus, for example, under no circumstances may the patent be interpreted as allegedly not providing a written description for a claim on the assertion that the precise wording of the claim is not set forth in haec verba in written description portion of the patent.

[0289] While the invention has been described in conjunction with the detailed description including but not limited to the embodiment statements thereof, the foregoing description is intended to illustrate and not limit the scope of the invention, which is defined by the scope of the appended claims. Thus, from the foregoing, it will be appreciated that, although specific nonlimiting embodiments of the invention have been described herein for the purpose of illustration, various modifications may be made without deviating from the scope of the invention. Other aspects, advantages, and modifications are within the scope of the following claims and the present invention is not limited except as by the appended claims.

What is claimed is:

1. A plastic extension for covering a nail plate comprising:

- a) a top surface, a bottom surface, a cuticle end, a distal end, a left edge and a right edge;
- b) a longitudinal curve running from the cuticle end to the distal end which increases in slope from the cuticle end to an apex and then decreases in slope from the apex to the distal end, and a latitudinal curve extending between the left and right edges and running from the cuticle end to the distal end;
- c) a longitudinal length of from about 15 mm to about 37 mm as a straight line from the midpoint of cuticle end to the midpoint of the distal end and running along a

presumed horizontal plane between the left edge and right edge which longitudinal length is divided into categories of short, medium and long;

- d) a width between left and right edges of from about 6 mm to about 15 mm wherein the width is measured at the widest distance between the left and right edges;
- e) an apex point on the longitudinal length which is directly coincident with and below the apex of the longitudinal curve wherein the length between the cuticle end of the longitudinal length and the apex point is the apex length;
- f) a ratio of the apex length to the longitudinal length selected from a range of from 0.27 to 0.49 which ratio determines the apex length according to an inverse relationship of the ratio and the longitudinal length;
- g) an apex height of from about 3 mm to about 6 mm as measured vertically from the top surface at the apex to the presumed horizontal plane between the left and right edges positioned directly below the apex;
- h) a cuticle height of from about 1.5 mm to about 3.5 mm relative to the presumed horizontal plane between the left and right edges;
- i) a ratio of the cuticle height to apex height of about 0.5 to about 0.62;
- j) a distal end thickness of no more than about 1 mm;
- k) a cuticle end thickness of no more than about 1 mm;
- l) an apex thickness of no more than about 1.5 mm.

2. The plastic extension according to claim 1 wherein the longitudinal length is categorized as short ranging from about 15 mm to about 21 mm, medium ranging from about 19 mm to about 28 mm and long ranging from about 24 mm to about 37 mm and wherein the ratio of the apex length to the longitudinal length is determined according to Table B:

TABLE B

Length Category	Length in mm	Apex to Length Ratio
Short	15-21	0.41-0.49
Medium	19-28	0.32-0.40
Long	24-37	0.27-0.31

3. The plastic extension according to claim 1 or 2 having a distal end height of no more than about 4.5 mm, preferably about 0.5 mm to about 4 mm, more preferably about 0.5-1 mm to about 3.5 mm.

4. The plastic extension according to any of the preceding claims wherein the distal end thickness is from about 0.6 mm to about 0.9 mm; the cuticle end thickness is from about 0.2 mm to about 0.4 mm; and the apex thickness is from about 0.6 mm to about 0.7 mm.

5. The plastic extension according to any of the preceding claims which does not move or rock when placed on a flat surface with the left and right edges of the plastic extension in contact with the flat surface and a downward force is applied to the cuticle end or the distal end of the plastic extension.

6. The plastic extension according to any of the preceding claims wherein the apex height is about 3.2 mm to about 5.7 mm and the ratio of cuticle height to apex height is about 0.52 to 0.62.

7. The plastic extension according to any of the preceding claims wherein the distal end height is about 1 mm to about 3.5 mm and the distal end thickness is about 0.61 mm to about 0.89 mm.

8. The plastic extension according to any of the preceding claims wherein the cuticle end thickness is about 0.21 mm to about 0.39 mm and the apex thickness is about 0.61 mm to about 0.7 mm.

9. The plastic extension according to any of the preceding claims wherein the cuticle end has a flex resistance of less than 350 g/mm, preferably less than 300 g/mm as measured according to a procedure employing a mechanical pressure tester having a vertical, mechanically operated pressure bar and opposing anvil wherein the plastic extension is arranged on the anvil so that the pressure bar contacts the cuticle end and the pressure bar is mechanically operated to apply pressure in grams per mm.

10. The plastic extension of any of the preceding claims wherein the plastic extension is placed on a fingernail of a user and the cuticle end of the plastic extension is compressed by a single finger to cause the cuticle end of the plastic extension to contact the corresponding cuticle surface of a fingernail.

11. The plastic extension according to any of the preceding claims further comprising one or more members of a group of plastic extensions of one overall shape wherein each member has a width and a length selected according to the dimensions of the thumbnail and fingernails of a hand onto which the members of the group are to be placed.

12. A nail applique kit comprising one or more plastic extensions according to any of the preceding claims and an adhesive.

13. The nail applique kit according to claim **12** wherein the adhesive is a UV curable gel, a cyanoacrylate adhesive,

a free radical curable composition, a pressure sensitive adhesive or a dual PSA coated slip.

14. The nail applique kit according to claim **13** wherein the adhesive is a UV curable gel.

15. The nail applique kit according to claim **14** wherein the nail applique kit further comprises a UV lamp preferably constructed of LED diodes for generation of electromagnetic radiation in the range of 280 to 420 nm.

16. The nail applicator kit according to claim **15** wherein the UV lamp provides selective control of the generated UV light wavelength and is capable of producing a flash irradiation of 340-360 nm or an in depth irradiation of 380-420 nm.

17. The nail applicator kit according to claim **16** wherein the flash irradiation is capable of producing a tacky UV gel state and an the in depth irradiation is capable of producing a cured UV gel.

18. A method using the nail applique kit of any of claims **12-17** comprising applying the adhesive to a fingernail nail plate, optionally processing the adhesive to render it at least partially adherent, fitting the plastic extension to the fingernail with adhesive, allowing the adhesive to bind the plastic extension to the fingernail nail plate.

19. Use of the nail applique kit according to any of claims **12-17**.

20. Use of the plastic extension of any of claims **1-11** as a fingernail applique on a fingernail.

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